Adaptive Threshold Estimation

Minimax laws, unbiased estimation, and a rare-event phase transition

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Abstract

We estimate the fraction $\theta = \mathbb{P}(P > 1/2)$ of a population of latent Bernoulli coins, charging one unit for each observed flip. Algorithms may create, revisit, and interleave coins and stop under a uniform expected budget. For any law μ of $D = |P-1/2|$ on $(0, 1/2]$, the minimax RMS error is comparable to $\omega_\mu(B^{-1/2})$, where $\omega_\mu(h) = \mathbb{E}_\mu \min\{1, h/D\}$. One algorithm depending only on the budget attains this rate for every μ. Unbiased estimation with uniformly finite variance and expected work is possible exactly when $I_\mu = \mathbb{E}_\mu D^{-1} < \infty$; the optimal uniform variance–work product lies between $I_\mu^2/16$ and $713 I_\mu^2$.

With the additional global rarity constraint $\theta \le \rho$, the variational quantity $\Psi_{\mu,\rho}$ gives a general lower bound and has an explicit optimizer. It characterizes the power-margin regimes below and above exponent two, but fails at the boundary. For the fixed quadratic law $F(d) = (4d)^2$, put $L = 1 + \log(1/\rho)$. A paper proof gives the sharp RMS order $\min\{\rho, \sqrt{\rho L(1 + \log L)/B}\}$, whereas Ψ gives only $\min\{\rho, \sqrt{\rho L/B}\}$. The extra factor comes from sign testing across hidden difficulty scales. The lower argument obtains one hard population pair for all capped single-coin policies and then transfers it to interacting population algorithms.

The original RMS lower bound, the untruncated 713 construction, the variational optimizer, and the finite-gap sign-entropy theorem are Lean-checked in their stated models. The original distribution-blind upper bound, unbiased converse, and full critical rare-event theorem remain paper proofs. This manuscript consolidates their statements, proofs, dependencies, and verification boundaries.

Research draft; no independent specialist review of the consolidated manuscript. Agent review and formal verification have different scopes. Priority and practical superiority are not established.

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1 Introduction and contribution

Estimating the fraction of a population lying above a latent threshold is not the same task as classifying every member of that population. Observations can be concentrated on promising or ambiguous members, and biased individual classifications can be combined through telescoping estimators. These possibilities make algorithm-independent lower bounds important.

We study the following basic observation model. A fresh coin has a latent probability P drawn from a population distribution G. Conditional on P, its observations are independent Bernoulli trials. An algorithm may adapt to its entire history: it can create new coins, interleave observations of old coins, and stop at a random time. Its target is

$$\theta(G) = \mathbb{P}_G(P > 1/2). \tag{1}$$

One coin flip is one unit of work. The principal obstruction is the population's mass near $1/2$, but a single power exponent need not describe that mass accurately.

Distributional and distribution-blind optimality. The central object is the entire function

$$\omega_\mu(h) = \int \min\{1, h/d\}\, \mu(\mathrm{d}d), \qquad \mu = \mathcal{L}(|P - 1/2|). \tag{2}$$

The main theorem identifies this function as the minimax error law. Thus irregular, discrete, and logarithmically modified margin distributions are covered by one statement, rather than by separate exponent calculations. The same algorithm attains the law for every magnitude distribution, rather than requiring a distribution-dependent allocation. A second theorem characterizes unbiased feasibility by an inverse moment and proves a sharp universal lower constant for the variance–work product. Writing $F_\mu(h) = \mathbb{P}(D \le h)$ makes the two parts of the modulus explicit:

$$\omega_\mu(h) = F_\mu(h) + h\mathbb{E}_\mu[D^{-1}; D > h].$$

The first term represents unresolved near-threshold mass; the second records the accumulated contribution of the remaining difficulty scales.

The lower bound makes many difficulty scales jointly hard: each scale contributes to the target separation, while every next observation has uniformly bounded information. The upper bound is a fixed geometric randomization of capped sequential classifiers. Its conditional second moment and expected work are both controlled by the same truncated inverse margin. This common conditional bound is what permits arbitrary margin laws. An exact two-success inverse-binomial calibration then chooses the number of production samples without estimating either their mean cost or variance. The underlying inverse-sampling identity is classical [8]; its role here is to eliminate all margin knowledge from the budget-indexed rule.

1.1 Relationship to established work

Nested simulation. Uniform inner-sampling analyses include Lee–Glynn [10] and Gordy–Juneja [11]; Broadie, Du, and Moallemi [9] also identify Lee's 1998 thesis as an early analysis and introduce sequential inner allocation for loss-probability estimation. Giles and Haji-Ali [2] combine adaptive inner sampling and multilevel Monte Carlo (MLMC) to attain $O(\varepsilon^{-2}|\log\varepsilon|^2)$ work for thresholded nested expectations. Haji-Ali, Spence, and Teckentrup [12] develop a broader adaptive MLMC framework for probabilities. We do not claim these methods or the critical upper rate as new. Our observed-state construction in proposition 19 verifies Assumptions 2.1 and 2.4 of [2] literally and proves the matching lower rate on an unstructured

subclass. It does not address models with additional smooth cross-state structure or alternative information oracles. Giles's survey [13] discusses the high kurtosis of level corrections for discontinuous functions; our confidence amplification uses only second moments, not an empirical variance estimate or a normal approximation.

Adaptive coin mixtures and information bounds. Lee and Valiant [1] prove fully adaptive optimal bounds for mixtures separated from $1/2$ by a fixed gap, including dependence on the positive fraction. Our gap-class specialization agrees with their rate in the balanced regime (corollary 15). The lower bounds here are driven by sign priors near $1/2$; this explains why a uniform local posterior inequality and the stopped KL chain rule suffice. Their more refined rare-event analysis is not reproduced by our supremum over all sign functions. The change-of-measure step is related to the standard sequential information inequality of Kaufmann, Cappé, and Garivier [14]; it is not a new general principle. Lee–Valiant also optimize estimators for known positive and negative conditional bias laws under a per-coin depth constraint. Here the magnitude law is arbitrary, the sign function is unrestricted, and our lower bounds have no per-coin depth constraint.

Randomization and unbiased estimation. Single-term multilevel debiasing is established methodology [3, 4]. Blanchet and Glynn [15] obtain finite work-normalized variance under regularity such as local twice differentiability of the outer function. Syed and Wang [5] study repeatedly nested expectations under suitable regularity conditions. Our threshold indicator is discontinuous: the inverse-margin condition and all-algorithm converse characterize what remains possible in this setting. The inverse-binomial sampling identity used for cost calibration dates to classical work of Haldane [8]; we give a self-contained proof and do not claim that identity as new.

Other observation models and scope. Brennan, Vinayak, and Jamieson [6] estimate mass above effect-size thresholds from noisy test statistics, rather than unrestricted sequential revisiting. Chen, Naslidnyk, and Briol [7] study nested expectations with kernel quadrature and identify minimax lower bounds as a research direction. Their smooth-kernel setting differs from ours. The proposed contribution is the arbitrary-margin minimax characterization, a distribution-blind optimal-rate algorithm, and the inverse-margin unbiased converse in the precisely specified experiment. The original arbitrary-margin theorem is not an instance-optimal or rare-event theorem. Section 11 treats a global rarity constraint separately; no general solution to all nested-expectation problems is claimed.

Rare events and the verification boundary. With a known upper bound $\theta \le \rho$, positive mass may concentrate on the most difficult magnitudes. The power-margin rates below and above exponent two are matched by different constructions. At the fixed quadratic law, a scale-entropy cost contributes an extra $\sqrt{1 + \log(1 + \log(1/\rho))}$ beyond the proposed variational modulus. Section 11 states the current results together; the full rare-event proofs appear in the appendices. This corrects the earlier matching-Ψ conjecture rather than assuming it.

The original RMS lower bound is proved in Lean for the actual hidden-magnitude experiment. The untruncated 713 estimator, variational optimizer, and finite-gap sign-entropy theorem have separate formal specifications. The matching budget-only upper algorithm, unbiased converse, and full critical theorem remain paper proofs. Section 12 records the exact scope and links.

2 Model, notation, and main results

Fix $a = 1/2$. Let μ be any probability measure on $(0, a]$; in particular, μ may have atoms, gaps, or a singular continuous part. A measurable function $q : (0, a] \to [0, 1]$ specifies a population by

$$D \sim \mu, \qquad S \mid D = d \sim \mathrm{Ber}(q(d)), \qquad P = \tfrac{1}{2} + (2S - 1)D. \tag{3}$$

The variables attached to distinct new coins are independent. The population class is $\mathcal{G}(\mu) = \{G_{\mu,q} : q \text{ measurable}\}$ and $\theta(\mu, q) = \int q(d)\, \mu(\mathrm{d}d)$. Individual magnitudes and signs are not observed. The upper algorithm does not know μ either. The distribution-wise minimax benchmark may grant knowledge of μ, so attaining it simultaneously without that knowledge is the stronger statement. The lower bound remains valid with still more information. The endpoint $D = 1/2$ is allowed and corresponds to deterministic coins; $D = 0$ is excluded.

An algorithm has an independent random seed, creates finitely many coins before each observation, selects one of its coins, and receives its next Bernoulli observation. It stops after T observations and returns a real number $\widehat{\theta}$. Infinite expected work is permitted in the model, but algorithms satisfying a finite uniform budget necessarily stop almost surely. Finite internal computation between observations is implicit. We count observations, not arithmetic operations or independent random bits. Every coin actually created by the upper algorithm receives at least one observation. Charging a fixed additional amount per new outer draw therefore changes its cost by at most a constant factor.

Define

$$R_\mu(B) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \sup_q \big(\mathbb{E}_{\mu,q}(\widehat{\theta} - \theta)^2\big)^{1/2}, \tag{4}$$

$$r_\mu(B, \delta) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \inf\{r \ge 0 : \sup_q \mathbb{P}_{\mu,q}(|\widehat{\theta} - \theta| > r) \le \delta\}. \tag{5}$$

These are expected-budget criteria, not deterministic maximum-work criteria.

For $h > 0$, write

$$A_\mu(h) = \mathbb{E}_\mu \min\{h^{-1}, D^{-1}\}, \qquad \omega_\mu(h) = hA_\mu(h), \qquad I_\mu = \mathbb{E}_\mu[D^{-1}] \in [2, \infty]. \tag{6}$$

The modulus is increasing and concave, $\omega_\mu(0+) = 0$, and $\omega_\mu(h) = 1$ for $h \ge a$, while $\omega_\mu(h) \ge \min\{1, 2h\}$. In particular, for $c \ge 1$ and $0 < b \le 1$,

$$\omega_\mu(ch) \le c\omega_\mu(h), \qquad \omega_\mu(bh) \ge b\omega_\mu(h). \tag{7}$$

Theorem 1 (Distribution-blind arbitrary-margin minimax law)**.** *There are universal constants $0 < c < C < \infty$ such that for every probability measure μ on $(0, 1/2]$, every $B \ge 1$, and every $0 < \delta \le 1/4$,*

$$c\omega_\mu(B^{-1/2}) \le R_\mu(B) \le C\omega_\mu(B^{-1/2}), \tag{8}$$

$$c\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right) \le r_\mu(B, \delta) \le C\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right). \tag{9}$$

More strongly, for each B there is one algorithm $\mathcal{A}_B$, depending only on B, that simultaneously for all μ, q satisfies

$$\mathbb{E}_{\mu,q} T \le B, \qquad \big(\mathbb{E}_{\mu,q}(\widehat{\theta} - \theta)^2\big)^{1/2} \le C\omega_\mu(B^{-1/2}). \tag{10}$$

There is likewise one algorithm depending only on (B, δ) for the upper confidence bound. Neither algorithm observes D or knows μ. In the lower bounds one may take $c = 1/64$ for RMS and $c = 15/(256\sqrt{2})$ for confidence. The lower bounds hold even if every coin's magnitude is revealed for free, or if the algorithm selects the magnitude of each fresh coin. In the latter experiment q is defined on all $(0, 1/2]$ and the target is still $\int q\, \mathrm{d}\mu$.

Formal scope. The RMS lower bound in the ordinary hidden-magnitude experiment is proved by `minimax_lower_bound`. The matching upper bound, confidence statement, and stronger revealed- or selected-magnitude claims retain paper-proof status.

This is distribution-wise minimax optimality: the worst case is over all sign functions at a fixed magnitude law. It is not instance optimality for each particular pair (μ, q). The budget is expected work, not an almost-sure cap. The modulus in the guarantee is evaluated at the true, possibly unknown distribution; it is not an observable confidence radius.

Theorem 2 (Exact unbiased feasibility and variance–work scale)**.** *An estimator unbiased for every $G \in \mathcal{G}(\mu)$ can have both*

$$V = \sup_q \mathrm{Var}_{\mu,q}(Y) < \infty, \qquad W = \sup_q \mathbb{E}_{\mu,q} T < \infty$$

if and only if $I_\mu < \infty$. When $I_\mu < \infty$,

$$\frac{1}{16} I_\mu^2 \;\le\; \inf_{Y \text{ unbiased}} VW \;\le\; 713 I_\mu^2, \tag{11}$$

where the infimum is over estimators with both uniform bounds finite. A single estimator, using neither μ nor any margin parameters, attains the upper bound simultaneously for every such magnitude law. It satisfies $\sup_q \mathbb{E}Y^2 \le 23 I_\mu$ and $\sup_q \mathbb{E}T \le 31 I_\mu$. The lower coefficient $1/16$ is the exact universal constant: equality is attained for every fixed-gap law $\mu = \delta_d$. Equality is not asserted for arbitrary nondegenerate μ.

Verification: the converse and sharp lower coefficient remain paper proofs. The constructive upper bound has the conditional-kernel and separate-family Lean coverage stated in section 7.

2.1 Unknown magnitude laws under an envelope

Let $F_\mu(t) = \mu((0,t])$, extended to equal one above a, and define

$$\mathcal{G}_{\preceq \mu} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \le t) \le F_\mu(t) \text{ for every } t > 0\}. \tag{12}$$

Here the actual magnitude law is unknown on $(0, 1/2]$. The inequality means its magnitudes are stochastically no smaller than those under μ. It also rules out an atom at $P = 1/2$.

Corollary 3 (Stochastic-dominance envelope version)**.** *Both conclusions of theorem 1 hold with the supremum taken over $\mathcal{G}_{\preceq \mu}$, with the same universal constants. The existence criterion and variance–work product bounds of theorem 2 also hold for this class. The finite-budget and unbiased algorithms need not know the envelope; it is used only to express a uniform guarantee.*

What distribution-blindness does not supply. The algorithm is specified without μ or an envelope. A numerical choice of budget for a prescribed accuracy, however, generally requires a bound on the unknown modulus. Neither the adaptive risk guarantee nor the unbiased estimator automatically supplies a data-computable confidence radius.

Corollary 4 (Critical nested-simulation obstruction)**.** *Let $U \sim \mathrm{Unif}[0,1]$, $X \in \{-1, 1\}$, and $D(U) = |\mathbb{E}[X \mid U]|/2$. Consider all measurable conditional models with $D(U) \sim \mathrm{Unif}(0, 1/4]$, with no cross-state regularity. For estimating $\mathbb{P}(\mathbb{E}[X \mid U] > 0)$, the minimax expected observation cost at RMS error ε, or at error ε with a fixed failure probability $\delta_0 = 1/16$, is*

$$\asymp \varepsilon^{-2} \log^2(1/\varepsilon), \qquad \varepsilon \downarrow 0.$$

The lower bound allows arbitrary selected-state queries and revisiting. Every model satisfies Assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with uniform constants.

The continuous-magnitude construction and proof are given in proposition 19. This is a worst-case sample-oracle statement on a subclass satisfying those assumptions, not a claim about every structured simulator or every possible source of information.

3 A history-uniform adaptive information bound

For Bernoulli probabilities $u, v \in (0,1)$,

$$\mathrm{kl}(u,v) \le \frac{(u-v)^2}{v(1-v)}. \tag{13}$$

This follows from $\log x \le x-1$ applied to the two likelihood ratios. We next control how an arbitrary local observation history changes two symmetric sign priors.

Lemma 5 (Posterior contraction at every magnitude)**.** *Fix $0 < d \le \frac{1}{2}$ and $0 \le \eta < \frac{1}{2}$. Under two experiments the sign priors are $q_\pm = \frac{1}{2} \pm \eta$, while the observation likelihoods conditional on the sign are the same. Let $L \in [0,\infty]$ be the likelihood ratio of an arbitrary local history of one coin under positive versus negative sign, let $\rho_\pm$ be its posterior sign probabilities, and let $p_\pm = \frac{1}{2} - d + 2d\rho_\pm$ be the next predictive success probabilities. If $0 < L < \infty$, then*

$$\rho_+ - \rho_- = \frac{2\eta L}{L + (\frac{1}{4} - \eta^2)(L-1)^2} \le 2\eta. \tag{14}$$

In every case,

$$\mathrm{kl}(p_+, p_-) \le \frac{32\, d^2\eta^2}{(1-4\eta^2)^2}. \tag{15}$$

The same bound holds for $\mathrm{kl}(p_-, p_+)$. The constant 32 is sharp: at $L = 1$ the ratio of the two sides of (15) tends to one as $\eta \downarrow 0$.

Proof. If $L \in \{0,\infty\}$, then $\rho_+ = \rho_- \in \{0,1\}$, so $p_+ = p_-$ and the divergence vanishes. This case cannot occur for $d < \frac{1}{2}$; for $d = \frac{1}{2}$ it occurs after the first observation. Now let $0 < L < \infty$ and write $a = \frac{1}{2} + \eta$, $b = \frac{1}{2} - \eta$. Then $a + b = 1$ and $ab = \frac{1}{4} - \eta^2 = (1-4\eta^2)/4$. Bayes' formula gives $\rho_+ = aL/(b+aL)$ and $\rho_- = bL/(a+bL)$, so

$$\rho_+ - \rho_- = \frac{(a^2-b^2)L}{(b+aL)(a+bL)}, \qquad (b+aL)(a+bL) = ab(L-1)^2 + (a+b)^2 L.$$

This proves (14). Also

$$\rho_-(1-\rho_-) = \frac{abL}{(a+bL)^2}, \qquad \rho_+(1-\rho_+) = \frac{abL}{(b+aL)^2}.$$

For fixed $v \in (0,1)$, the function $u \mapsto \mathrm{kl}(u,v)$ vanishes together with its first derivative at $u = v$, and its second derivative is $1/\{u(1-u)\}$. Since $p_- \le p_+$, Taylor's theorem gives

$$\mathrm{kl}(p_+, p_-) \le \frac{(p_+ - p_-)^2}{2\min_{\xi \in [p_-, p_+]} \xi(1-\xi)}.$$

The same Taylor upper bound applies with the two arguments interchanged, since its right side depends only on their squared difference and the interval between them. Thus it controls both KL directions. Every such ξ equals $\frac{1}{2} - d + 2d\rho$ for some $\rho \in [\rho_-, \rho_+] \subset (0,1)$. Because $d \le \frac{1}{2}$,

$$\xi(1-\xi) = \tfrac{1}{4} - d^2(2\rho-1)^2 \ge \tfrac{1}{4}\{1 - (2\rho-1)^2\} = \rho(1-\rho).$$

By concavity, the minimum of $\rho(1-\rho)$ over $[\rho_-, \rho_+]$ is attained at an endpoint. Since $p_+ - p_- = 2d(\rho_+ - \rho_-)$, the preceding displays give

$$\mathrm{kl}(p_+, p_-) \le 2d^2 \max\Big\{\frac{(\rho_+ - \rho_-)^2}{\rho_-(1-\rho_-)}, \frac{(\rho_+ - \rho_-)^2}{\rho_+(1-\rho_+)}\Big\} = \frac{8d^2\eta^2}{ab} \max\Big\{\frac{L}{(b+aL)^2}, \frac{L}{(a+bL)^2}\Big\}.$$

The maps $L \mapsto L/(b+aL)^2$ and $L \mapsto L/(a+bL)^2$ are maximized at $L = b/a$ and $L = a/b$ respectively, both with value $1/(4ab)$. Therefore $\mathrm{kl}(p_+, p_-) \le 2d^2\eta^2/(ab)^2$, which is (15).

For sharpness, at $L = 1$ we have $p_\pm = \frac{1}{2} \pm 2d\eta$ and

$$\mathrm{kl}(p_+, p_-) = 4d\eta \log \frac{1+4d\eta}{1-4d\eta} = 32d^2\eta^2\{1 + O(d^2\eta^2)\}. \qquad \square$$

Fix $h > 0$ and take

$$\eta_h(d) = \frac{1}{8}\min\{1, h/d\}, \qquad q_\pm(d) = \frac{1}{2} \pm \eta_h(d). \tag{16}$$

The two targets are centered at $1/2$ and separated by

$$\Delta_h = \theta_+ - \theta_- = 2\mathbb{E}_\mu \eta_h(D) = \frac{1}{4}\omega_\mu(h). \tag{17}$$

Because $d\eta_h(d) \le h/8$, every next observation has predictive KL at most $128h^2/225$ in either direction.

Proposition 6 (Stopped-transcript bound)**.** *For every adaptive algorithm in any of the experiments of theorem 1, its stopped transcript laws satisfy*

$$\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}) \le \frac{128}{225} h^2 \mathbb{E}_+ T. \tag{18}$$

Consequently, if it estimates both targets to radius $\varepsilon < \Delta_h/2$ with failure probability at most $\delta < 1/2$,

$$\mathbb{E}_+ T \ge \frac{225}{128h^2}\,\mathrm{kl}(1-\delta, \delta). \tag{19}$$

Proof. First reveal all magnitudes; an algorithm not using them is a special case. Conditional on a realized transcript, action-selection and randomization factors are common to the two hypotheses. Conditional on the hypothesis and revealed magnitudes, the posterior of a coin's sign depends only on its own observations: the likelihood factorizes by coin. Thus the local posterior identity applies even after arbitrary revisiting and interleaving. For a fresh coin the same assertion has $L = 1$. At $d = 1/2$ the first observation has full support under both hypotheses, and after it the sign is revealed, so later observations contribute zero KL.

A newly drawn magnitude has the same law in both experiments and contributes zero KL. A selected magnitude is an algorithmic action and also contributes zero conditional KL. The chain rule, first stopped at $T \wedge m$, bounds transcript KL by $128h^2\mathbb{E}_+(T \wedge m)/225$. Padding the transcript after its stopping time and taking increasing finite prefixes gives (18) by monotone convergence of expected work and the finite-prefix characterization of relative entropy. If expected work is infinite, the bound is vacuous. Interchanging the hypotheses gives the reverse bound with $\mathbb{E}_- T$.

Thresholding $\widehat{\theta}$ at $1/2$ gives a test with both errors at most δ, since the estimation radius is strictly less than half the separation. Data processing gives $\mathrm{KL} \ge \mathrm{kl}(1-\delta, \delta)$, proving (19). $\square$

The strict separation in this proposition is intentional. At exactly half the separation, the constant output $1/2$ is within tolerance for both members of the pair, so a test cannot be inferred without a limiting argument over strictly separated pairs.

4 A universal capped classifier

This section does not assume a magnitude law. Fix a coin with $P = 1/2 + (2S-1)d$, where $0 < d \le 1/2$ and $S \in \{0,1\}$. For a dyadic cap $N = 2^\ell$, examine prefixes of lengths $1, 2, 4, \dots, N$. At a prefix $n < N$, stop when

$$\left|\overline{X}_n - \frac{1}{2}\right| > t_{n,N}, \qquad t_{n,N} = \sqrt{\frac{1 + \log(N/n)}{n}}, \tag{20}$$

and output the side of $1/2$ occupied by the sample mean. If no early stop occurs, output $Z_N = \mathbf{1}\{\overline{X}_N > 1/2\}$; a final tie is negative. Let τ_N be the number of observations used.

Set

$$m_N(d) = \min\{N, d^{-2}[1 + \log_+(Nd^2)]\}, \qquad \log_+ x = \max\{0, \log x\}. \tag{21}$$

Lemma 7 (Conditional error and work)**.** *For every dyadic $N \ge 1$, both signs, and $0 < d \le 1/2$,*

$$e_N(d,S) := \mathbb{P}(Z_N \ne S \mid d, S) \le \min\{1, (Nd^2)^{-2}\}, \tag{22}$$

$$\mathbb{E}[\tau_N \mid d, S] \le 17 m_N(d). \tag{23}$$

Proof. A wrong early stop at a fixed prefix n requires a one-sided deviation of at least $d + t_{n,N}$. Hoeffding's inequality and a union bound over fixed prefixes, not a conditioning on the stopping event, give

$$\begin{aligned} e_N(d,S) &\le e^{-2Nd^2} + \sum_{\substack{n<N \\ n \text{ dyadic}}} e^{-2n(d+t_{n,N})^2} \\ &\le e^{-2Nd^2} + e^{-2} \sum_{\substack{n<N \\ n \text{ dyadic}}} (n/N)^2 e^{-2nd^2}. \end{aligned} \tag{24}$$

For every dyadic sequence of positive u values and every $v, b > 0$,

$$\sum_u u^v e^{-bu} \le \frac{\Gamma(v+1)}{(1-2^{-v})b^v}. \tag{25}$$

Indeed, the disjoint intervals $[u/2, u]$ satisfy $\int_{u/2}^u y^{v-1} e^{-by}\,\mathrm{d}y \ge e^{-bu} u^v (1-2^{-v})/v$. With $x = Nd^2 \ge 1$, the sum in (24) is bounded by $x^{-2}(2/3)$, and $x^2 e^{-2x} \le e^{-2}$. Hence the right side is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. The bound by one handles $x < 1$.

The dyadic survival identity and the fact that a surviving prefix cannot have crossed its threshold imply

$$\begin{aligned} \mathbb{E}\tau_N &= 1 + \sum_{n<N,\, n \text{ dyadic}} n\mathbb{P}(\tau_N > n) \\ &\le 1 + \sum_{n<N,\, n \text{ dyadic}} n\{\mathbf{1}[d \le 2t_{n,N}] + 2e^{-nd^2/2}\}. \end{aligned} \tag{26}$$

For $x \ge 1$, the inequality $d \le 2t_{n,N}$ implies $nd^2 \le 4(1 + \log x)$: if $nd^2 < 1$ this is immediate, and otherwise $\log(N/n) = \log(x/(nd^2)) \le \log x$. The corresponding dyadic sum of n is at most $8d^{-2}(1 + \log x)$. Using (25) with $v = 1, b = 1/2$ bounds the exponential part of (26) by $8d^{-2}$. Since $1 \le d^{-2}$, the total is at most $17d^{-2}(1 + \log x)$. Also $\tau_N \le N$ deterministically, and $1 + \log x \le x$ for $x \ge 1$. For $x < 1$, $m_N(d) = N$, completing the proof. □

4.1 One geometric level law for every population

On a common coin stream define

$$\Delta_0 = Z_1, \qquad \Delta_\ell = Z_{2^\ell} - Z_{2^{\ell-1}} \quad (\ell \geq 1). \tag{27}$$

A coupled difference is computed by running both classifiers on the same prefix until both have stopped. Its work W_ℓ is the maximum of their stopping times and hence at most their sum. Let

$$r = 2^{-1/2}, \qquad p_\ell = (1-r)r^\ell, \qquad \mathbb{P}(J = \ell) = p_\ell. \tag{28}$$

Choose J independently of a fresh coin and its observations. For a finite cutoff L, return

$$Y_L = \frac{\Delta_J}{p_J}\mathbf{1}\{J \leq L\}. \tag{29}$$

When $J > L$, no coin is sampled and the output and work are both zero. Write $\mathcal{W}_L$ for actual work. For $L = \infty$, drop the indicator and write $Y, \mathcal{W}$.

Lemma 8 (Truncated inverse-margin control)**.** *For every finite $L \geq 0$, $N = 2^L$, both signs, and $0 < d \leq 1/2$,*

$$\mathbb{E}[Y_L \mid d, S] = \mathbb{E}[Z_N \mid d, S], \tag{30}$$

$$\mathbb{E}[Y_L^2 \mid d, S] \leq 85 \min\{\sqrt{N}, d^{-1}\}, \tag{31}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \leq 149 \min\{\sqrt{N}, d^{-1}\}. \tag{32}$$

Moreover,

$$\sum_{\ell \geq 0} \mathbb{E}[|\Delta_\ell| \mid d, S] \leq C_0[1 + \log(1/d)] \tag{33}$$

for a universal C_0. Whenever $\mathbb{E}\log(1/D) < \infty$, Y is integrable and unbiased for θ. This logarithmic condition is sufficient here; no necessity claim for integrability alone is made. If $\mathbb{E}D^{-1} < \infty$, its second moment is at most $85\mathbb{E}D^{-1}$ and its expected work at most $149\mathbb{E}D^{-1}$.

Proof. The finite expectation telescopes. For $\ell \geq 1$, a nonzero difference requires at least one classifier to be wrong, so, with $N_\ell = 2^\ell$,

$$\mathbb{E}[\Delta_\ell^2 \mid d, S] \leq 5 \min\{1, (N_\ell d^2)^{-2}\}, \qquad \mathbb{E}[W_\ell \mid d, S] \leq 34 m_{N_\ell}(d). \tag{34}$$

The same loose inequalities cover $\ell = 0$. Consequently

$$\mathbb{E}[Y_L^2 \mid d, S] \leq \frac{5}{1-r} \sum_{\ell=0}^{L} \sqrt{N_\ell} \min\{1, (N_\ell d^2)^{-2}\}, \tag{35}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \leq 34(1-r) \sum_{\ell=0}^{L} N_\ell^{-1/2} m_{N_\ell}(d). \tag{36}$$

Split each sum at $N_\ell = d^{-2}$. Before that point, both summands are $\sqrt{N_\ell}$ and their sum is bounded by $(1-r)^{-1}\min\{\sqrt{N}, d^{-1}\}$. After that point, the second-moment sum is a geometric tail with ratio $2^{-3/2}$ and is at most $(1-r^3)^{-1}d^{-1}$. Its full constant is bounded by

$$\frac{5}{1-r}\left(\frac{1}{1-r} + \frac{1}{1-r^3}\right) < 85.$$

For work, the first N_ℓ after the split has $1 < N_\ell d^2 \leq 2$. Its tail is bounded by

$$d^{-1} \sum_{k \geq 0} r^k (1 + \log 2 + k \log 2).$$

Combining the prefix and tail gives the constant

$$34\left(2+\log 2+\frac{r\log 2}{1-r}\right)<149.$$

If the cutoff precedes the split, only the smaller prefix bound is needed. This proves (31)–(32).

Because $|\Delta_\ell|=\Delta_\ell^2$ for $\ell\geq 1$, the number of levels before the split is $O(1+\log(1/d))$, and the subsequent error bounds form a convergent geometric series. This proves (33). Absolute summability allows Fubini and telescoping to S, using (22). Averaging over (D,S) proves unbiasedness whenever the stated logarithmic moment is finite. The inverse-moment condition implies that condition. Finally take $L\to\infty$ in the nonnegative series for second moment and work. □

This is the standard single-term randomized multilevel identity [4]; the common conditional bounds and their distributional consequences are the key additional estimates. The estimator's output need not lie in $[0,1]$. The interior-magnitude obstruction to nonnegative output is given in section 6.1.

5 Proof of the minimax law

5.1 Exact calibration without a known margin law

For $h=N^{-1/2}$ define

$$b_\mu(h)=\mathbb{E}_\mu\min\{1,(h/D)^4\}\leq\omega_\mu(h). \tag{37}$$

One call of the cutoff estimator returns $(Y_L,\mathcal{W}_L)$ with

$$0\leq\mathcal{W}_L\leq N,\quad c:=\mathbb{E}\mathcal{W}_L\leq 149A_\mu(h),\quad v:=\mathrm{Var}(Y_L)\leq 85A_\mu(h),\quad |\mathbb{E}Y_L-\theta|\leq b_\mu(h). \tag{38}$$

Moreover $c>0$, because level zero has probability $1-r>0$ and costs one observation. The following wrapper handles the unknown c exactly.

Lemma 9 (Two-success cost calibration)**.** *Suppose an independent call returns (Z,C) with $0\leq C\leq N$, $c=\mathbb{E}C>0$, $\mathbb{E}Z=\zeta$, and $\mathrm{Var}(Z)=v<\infty$. Dependence of value and cost within a call is unrestricted. Run independent pilot calls and mark each with conditional probability C_i/N, using independent randomization. Stop at the second mark, at call K. Discard all pilot values and average $M=K-1$ fresh independent production values. If $\widehat{\zeta}$ is that average and T is the total pilot-plus-production cost, then*

$$\mathbb{E}\widehat{\zeta}=\zeta,\qquad \mathrm{Var}(\widehat{\zeta})=\frac{vc}{N},\qquad \mathbb{E}T=4N-c\leq 4N. \tag{39}$$

The procedure uses neither c nor v.

Proof. The pilot marks are independent Bernoulli variables with parameter $p=c/N>0$, so K has negative-binomial probabilities

$$\mathbb{P}(K=k)=(k-1)p^2(1-p)^{k-2},\quad k\geq 2,\qquad \mathbb{E}K=2/p.$$

In particular $K<\infty$ almost surely, and the classical inverse-binomial identity [8] is immediate:

$$\mathbb{E}\frac{1}{K-1}=p^2\sum_{k=2}^{\infty}(1-p)^{k-2}=p. \tag{40}$$

The event $\{K \geq i\}$ depends only on earlier calls and marks, so is independent of C_i. Tonelli's theorem yields

$$\mathbb{E}\sum_{i=1}^{K} C_i = \sum_{i\geq 1} \mathbb{P}(K \geq i)c = c\mathbb{E}K = 2N.$$

Production calls are independent of the pilot. Their mean total cost is $c\mathbb{E}(K-1) = 2N-c$, and conditional on K the production average has mean ζ and variance $v/(K-1)$. Total expectation and variance give (39). For $p = 1$, $K = 2$ deterministically and the same identities hold. □

5.2 Distribution-blind upper bounds

For $B \geq 4$, choose the largest dyadic $N \leq B/4$, set $L = \log_2 N$, and apply lemma 9 to $(Z, C) = (Y_L, \mathcal{W}_L)$. By (38),

$$\mathbb{E}T \leq 4N \leq B, \qquad \mathbb{E}(\widehat{\theta} - \theta)^2 \leq \omega_\mu(h)^2 + \frac{85 \cdot 149 A_\mu(h)^2}{N} = 12666\omega_\mu(h)^2. \tag{41}$$

Since $N > B/8$, (7) gives the RMS upper bound, for example with $C_R = \sqrt{8 \cdot 12666}$. For $1 \leq B < 4$, output $1/2$ with no observations; then $\omega_\mu(B^{-1/2}) = 1$. This proves (10) without any knowledge of μ or q.

For confidence, choose the least odd integer $k \geq (32/9)\log(1/\delta)$; then $k < 5\log(1/\delta)$ for $0 < \delta \leq 1/4$. If $B/k \geq 4$, use k independent groups, each with budget B/k, and return their median. A group's error exceeds $\sqrt{8}C_R\omega_\mu(\sqrt{k/B})$ with probability at most $1/8$, by its mean-square bound. Hoeffding's inequality bounds the probability that at least half the groups fail by $e^{-9k/32} \leq \delta$. Costs add in expectation, and

$$\omega_\mu(\sqrt{k/B}) \leq \sqrt{5}\,\omega_\mu\left(\sqrt{\log(1/\delta)/B}\right).$$

If $B/k < 4$, output $1/2$; then $\sqrt{\log(1/\delta)/B} > 1/\sqrt{20}$, and $\omega_\mu(\sqrt{\log(1/\delta)/B}) > 1/\sqrt{5}$. This handles the remaining case with a universal constant.

Implementation and scope. Pilot values are discarded; recycling them into the final average would invalidate the independence calculation. Value and cost may be correlated within any call. The random pilot and production counts are unbounded, so the guarantee concerns expected observations, not a run-by-run work cap. Projection of the final finite-budget estimate onto $[0, 1]$ can only reduce its error, but is not used in the variance identity.

5.3 Matching lower bounds

For RMS error use (16) with $h = (4\sqrt{B})^{-1}$. By proposition 6, transcript KL is at most $8/225$, so Pinsker's inequality gives total variation at most $2/15$. Thresholding an estimator at $1/2$ gives a test whose error probabilities sum to at most $8R^2/\Delta_h^2$, if its RMS error under both hypotheses is at most R. Every test has error sum at least $1 - 2/15 > 1/2$. Hence $R > \Delta_h/4 = \omega_\mu(h)/16 \geq \omega_\mu(B^{-1/2})/64$. Taking the infimum proves the stated weak inequality.

For confidence write $K_\delta = \mathrm{kl}(1-\delta, \delta)$ and take

$$h = \frac{15}{16\sqrt{2}}\sqrt{K_\delta/B}.$$

Then $225K_\delta/(128h^2) = 4B > B$; by proposition 6 no admissible estimator has radius strictly smaller than $\Delta_h/2 = \omega_\mu(h)/8$. On $0 < \delta \leq 1/4$,

$$K_\delta = (1-2\delta)\log\frac{1-\delta}{\delta} \geq \frac{1}{4}\log(1/\delta). \tag{42}$$

Indeed $1-2\delta \geq 1/2$ and $1-\delta \geq \sqrt{\delta}$. Thus $h \geq 15\sqrt{\log(1/\delta)/B}/(32\sqrt{2})$. Apply (7) to obtain the coefficient $15/(256\sqrt{2})$. This completes the proof of theorem 1.

5.4 Proof for a stochastic-dominance envelope

The functions $d \mapsto \min\{1, h/d\}$, $d \mapsto \min\{1, (h/d)^4\}$, and $d \mapsto \min\{h^{-1}, d^{-1}\}$ are decreasing. Their expectations for any magnitude law in (12) are therefore bounded by their expectations under μ. The distribution-blind upper algorithm is unchanged; its risk bound is no larger than the envelope value. The lower bound follows by restricting to $\mathcal{G}(\mu)$, which is a subclass. The same reasoning with d^{-1} and theorem 2 proves corollary 3.

Remark 10 (Why the statement uses error as a function of budget)**.** For highly irregular μ, inverting ω_μ at two different constant multiples of ε need not preserve multiplicative constants in work. Statements such as $B \asymp \omega_\mu^{-1}(\varepsilon)^{-2}$ need a regularity qualification or explicit constants inside the inverse. The modulus bounds (8)–(9) do not have this defect. Polynomial and logarithmic examples below can be inverted with the stated asymptotic orders.

6 Proof of the unbiased characterization

Sufficiency already follows from lemma 8. The improved $713I_\mu^2$ bound uses the modified classifier proved in section 7. We prove the converse using only two-point inequalities and the stopped KL chain rule; no differentiation of an estimator's expectation is needed.

Lemma 11 (Two-point variance inequality)**.** *Let $\mathbb{P}, \mathbb{Q}$ be probability measures on a common space, let $H^2 = 1 - \int \sqrt{d\mathbb{P}\, d\mathbb{Q}}$, and let Y be a real statistic with finite second moment under both. Then $\Delta = \mathbb{E}_\mathbb{P} Y - \mathbb{E}_\mathbb{Q} Y$ satisfies*

$$\Delta^2(1 - 2H^2) \leq 4H^2\big(\mathrm{Var}_\mathbb{P} Y + \mathrm{Var}_\mathbb{Q} Y\big), \tag{43}$$

Moreover, $2H^2 \leq \mathrm{KL}(\mathbb{P}, \mathbb{Q})$, and the symmetric bound is

$$8H^2 \leq \mathrm{KL}(\mathbb{P}, \mathbb{Q}) + \mathrm{KL}(\mathbb{Q}, \mathbb{P}). \tag{44}$$

Proof. Let p, q be densities with respect to $\nu = \mathbb{P} + \mathbb{Q}$, and let $c = (\mathbb{E}_\mathbb{P} Y + \mathbb{E}_\mathbb{Q} Y)/2$. Then $\Delta = \int (Y-c)(\sqrt{p}-\sqrt{q})(\sqrt{p}+\sqrt{q})\, d\nu$. We use three facts: $\int (\sqrt{p}-\sqrt{q})^2\, d\nu = 2H^2$; $(\sqrt{p}+\sqrt{q})^2 \leq 2(p+q)$; and $\mathbb{E}_\mathbb{P}(Y-c)^2 = \mathrm{Var}_\mathbb{P} Y + \Delta^2/4$, with the same identity under $\mathbb{Q}$. Together with the Cauchy–Schwarz inequality they give

$$\Delta^2 \leq \int (\sqrt{p} - \sqrt{q})^2\, d\nu \cdot 2 \int (Y-c)^2 (p+q)\, d\nu = 4H^2\Big(\mathrm{Var}_\mathbb{P} Y + \mathrm{Var}_\mathbb{Q} Y + \tfrac{1}{2}\Delta^2\Big).$$

Rearranging gives (43). Finally, by Jensen's inequality, $-\frac{1}{2}\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_\mathbb{P} \log\sqrt{q/p} \leq \log(1 - H^2) \leq -H^2$. For the symmetric inequality, use the pointwise bound

$$(x-y)\log(x/y) \geq 4(\sqrt{x} - \sqrt{y})^2, \qquad x, y > 0.$$

Indeed, putting $r = \sqrt{x/y}$, the difference is $2y(r-1)(r+1)g(r)$, where $g(r) = \log r - 2(r-1)/(r+1)$ has $g(1) = 0$ and $g'(r) = (r-1)^2/[r(r+1)^2] \geq 0$. Integrating the pointwise bound gives (44), with zero-density cases interpreted by limits. □

Converse in theorem 2. Suppose Y is unbiased for every q, with finite uniform variance V and expected work W. Fix $h > 0$ and $0 < t \leq h/8$, and consider

$$q_\pm(d) = \tfrac{1}{2} \pm t\varphi_h(d), \qquad \varphi_h(d) = \min\{h^{-1}, d^{-1}\}.$$

Then $\eta(d) = t\varphi_h(d) \le t/h \le 1/8$ and $d\eta(d) \le t$. Lemma 5 and the stopped-transcript argument give both

$$\max\{\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}), \mathrm{KL}(\mathsf{P}_-^{\mathcal{A}}, \mathsf{P}_+^{\mathcal{A}})\} \le \kappa_t := \frac{32t^2W}{(1-4t^2/h^2)^2}. \tag{45}$$

Each direction uses its own expected work, bounded by W. Unbiasedness gives mean separation $2tA_\mu(h)$. By (44), $H^2 \le \kappa_t/4$. Applying lemma 11 with both variances at most V yields

$$4t^2A_\mu(h)^2(1-\kappa_t/2) \le 2\kappa_t V, \qquad A_\mu(h)^2(1-\kappa_t/2) \le \frac{16VW}{(1-4t^2/h^2)^2}.$$

First let $t \downarrow 0$, with h fixed, giving $A_\mu(h)^2 \le 16VW$. Then $A_\mu(h) \uparrow I_\mu$ as $h \downarrow 0$ by monotone convergence. This proves $I_\mu < \infty$ and $VW \ge I_\mu^2/16$. The envelope class contains $\mathcal{G}(\mu)$, so the same lower bound holds there. □

Sharpness of the universal coefficient. For $\mu = \delta_d$, $0 < d \le 1/2$, one fresh observation $X \in \{0,1\}$ gives

$$Y = \frac{X - 1/2 + d}{2d}, \qquad \mathbb{E}Y = q(d), \qquad \sup_q \mathrm{Var}(Y) = \frac{1}{16d^2}, \qquad W = 1. \tag{46}$$

Since $I_\mu = 1/d$, equality holds. Hence $1/16$ cannot be increased in a bound valid for every magnitude law. This does not assert equality for every nondegenerate law, or that the parameter-free construction achieves the optimal constant for each law.

Uniformity matters. The impossibility is over the whole sign class at a fixed μ, or over the stated envelope class. It does not say that each single population with infinite inverse moment forbids every efficient unbiased estimator. For a particular known population, the target might already be known. Likewise, a small structured family of sign functions can be easier than an arbitrary measurable q.

6.1 Why a universal unbiased output must sometimes be negative

Proposition 12 (No nonnegative unbiased output except at a deterministic endpoint)**.** *If $\mu((0,1/2)) > 0$, no algorithm stopping almost surely for every q and always returning a nonnegative value is unbiased on $\mathcal{G}(\mu)$. For $\mu = \delta_{1/2}$ the obstruction disappears: one observation is an unbiased $[0,1]$-valued estimator with unit work.*

Proof. Under $q \equiv 0$ the target is zero, so nonnegativity and unbiasedness force the output to vanish almost surely. Compare this with $q_1(d) = \mathbf{1}\{d < 1/2\}$, whose target is $\mu((0,1/2)) > 0$. Interior coins give positive likelihood to every finite Bernoulli history under both models; endpoint coins have the same deterministic sign zero. Finite stopped transcript laws are equivalent. Intersecting with each $\{T = n\}$ and taking a countable union gives equivalence of the full stopped transcripts. Seeds and any revealed magnitudes have the same distribution. Thus the output is also zero almost surely under q_1, a contradiction. If $\mu = \delta_{1/2}$, the first observation is exactly the sign. □

This elementary support argument is not offered as a separate novelty claim. It explains the signed outputs in the construction. Clipping is allowed for a biased finite-accuracy estimator but invalidates exact unbiasedness.

7 A uniform variance–work upper constant of 713

The finite-budget construction in sections 4 and 5 uses the original threshold and constants 85, 149. For the unbiased upper bound in theorem 2, change the classifier threshold to

$$t_{n,N} = \sqrt{\frac{1/4 + \log(N/n)}{n}} \qquad (n < N),$$

keeping the same dyadic checkpoints, strict crossing rule, negative final tie convention, shared-prefix increments, and independent level law $p_\ell = (1-r)r^\ell$, $r = 2^{-1/2}$. In this section $Z_N, \tau_N, \Delta_\ell, Y_L, \mathcal{W}_L$ denote this modified construction only. It observes no magnitude or sign and uses no distributional parameters.

Theorem 13 (Separate conditional moment bounds)**.** *For every $0 < d \le 1/2$, either sign, and finite L, $N = 2^L$,*

$$\mathbb{E}[Y_L^2 \mid d, S] \le 23 \min\{\sqrt{N}, d^{-1}\}, \qquad \mathbb{E}[\mathcal{W}_L \mid d, S] \le 31 \min\{\sqrt{N}, d^{-1}\}.$$

If $I_\mu < \infty$, the untruncated estimator is square-integrable, unbiased, and has finite expected actual work, with

$$\sup_q \mathbb{E}_{\mu,q} Y^2 \le 23 I_\mu, \qquad \sup_q \mathbb{E}_{\mu,q} \mathcal{W} \le 31 I_\mu.$$

Consequently the product of the separate variance and work suprema is at most $713 I_\mu^2$.

Error envelope. An incorrect early stop at prefix n requires a one-sided deviation of at least $d + t_{n,N}$. Hoeffding's inequality and a union bound over the fixed dyadic prefixes yield, with $x = Nd^2$,

$$e_N \le e^{-2x} + e^{-1/2} N^{-2} \sum_{n<N,\ n \text{ dyadic}} n^2 e^{-2nd^2} \le e^{-2x} + \tfrac{2}{3} e^{-1/2} x^{-2}.$$

For the second step integrate ue^{-bu} over the disjoint intervals $[n/2, n]$: each integral is at least $3n^2 e^{-bn}/8$, so the sum is at most $8/(3b^2)$. Since $x^2 e^{-2x} \le e^{-2}$ for every $x > 0$,

$$e_N \le \min\{1, Ax^{-2}\}, \qquad A = e^{-2} + \tfrac{2}{3} e^{-1/2} < \tfrac{1}{7} + \tfrac{5}{12} = \tfrac{47}{84} < \tfrac{3}{5}.$$

Here $e^2 > 7$ and $e^{1/2} > 8/5$ follow from their positive power series. For $\ell \ge 1$, a nonzero increment requires a fine or coarse error; the coarse scale has $x/2$. Hence

$$\mathbb{E}\Delta_\ell^2 \le \min\{1, 5Ax^{-2}\} \le \min\{1, 3x^{-2}\}. \tag{47}$$

The same envelope covers $\ell = 0$, because $d^2 \le 1/4$ and $\Delta_0 = Z_1 \in \{0, 1\}$.

Second moment. Put $d' = d/3^{1/4}$. Summing (47) divided by p_ℓ and splitting at $2^\ell (d')^2 = 1$ gives

$$\mathbb{E}Y_L^2 \le C_0 \min\{\sqrt{2^L}, 1/d'\}, \qquad C_0 = \frac{1}{1-r}\left(\frac{1}{1-r} + \frac{1}{1-r^3}\right) < 17.$$

Before the split the terms are geometric with size $\sqrt{2^\ell}$; after it their ratio is r^3. This includes cutoffs before the split. The rational comparison $r < 442/625$ and monotonicity of the displayed expression prove $C_0 < 17$ by integer arithmetic. Since $3^{1/4} < 4/3$, the conditional constant is less than $68/3 < 23$.

Expected work. Take $c = 13/7$. Survival at a dyadic prefix $n < N$ and $d > ct_{n,N}$ requires a one-sided deviation larger than $(1 - 1/c)d$. Therefore

$$\mathbb{P}(\tau_N > n) \le \mathbf{1}\{d \le ct_{n,N}\} + \exp\{-2(1 - 1/c)^2 nd^2\}.$$

Suppose $x = Nd^2 \ge 1$ and write $u = nd^2$. The indicator implies $u \le c^2[1/4 + \log(x/u)]$. If $u < 1$, then $u \le 1 + c^2 \log x$; if $u \ge 1$, use $\log u \ge 0$ and $c^2/4 = 169/196 < 1$ for the same conclusion. Its dyadic contribution is at most $2d^{-2}(1 + c^2 \log x)$. For $b > 0$, integration on $[n/2, n]$ gives $\sum ne^{-bnd^2} \le 2/(bd^2)$. The survival identity and $1 \le d^{-2}/4$ now give

$$\mathbb{E}\tau_N \le d^{-2}\left[\frac{125}{18} + \frac{338}{49}\log x\right] \le 7d^{-2}(1 + \log x).$$

Together with $\tau_N \le N$, this proves $\mathbb{E}\tau_N \le 7m_N(d)$.

On a shared stream the cap-N threshold is at least the cap-$N/2$ threshold at every prefix before $N/2$. If the fine classifier stops there, the coarse one has already stopped; at or after $N/2$ the coarse one has reached its cap. Thus $\tau_{N/2} \le \tau_N$ pathwise and the actual work of the coupled increment equals τ_N. Level zero costs one. It follows that

$$\mathbb{E}\mathcal{W}_L \le 7(1 - r)\sum_{\ell=0}^{L} 2^{-\ell/2} m_{2^\ell}(d) \le 7\left(2 + \frac{\log 2}{1 - r}\right)\min\{\sqrt{2^L}, d^{-1}\}.$$

To obtain the last inequality, split at $2^\ell d^2 = 1$ as in lemma 8. The first tail value satisfies $1 < 2^\ell d^2 \le 2$; the remaining sum is bounded by $d^{-1}\sum_{k\ge 0} r^k(1 + \log 2 + k \log 2)$. Using $r < 71/100$ and $\log 2 < 7/10$ bounds the final constant by $896/29 < 31$.

Untruncated limit and separate suprema. Integrate the conditional bounds over (D, S). Monotone convergence of the nonnegative moment and cost series gives finite second moment and work when $I_\mu < \infty$. More explicitly,

$$\mathbb{E}[(Y - Y_L)^2] = \sum_{\ell > L} \frac{\mathbb{E}\Delta_\ell^2}{p_\ell} \longrightarrow 0.$$

Thus $Y_L \to Y$ in L^2 and L^1. Finite telescoping gives $\mathbb{E}Y_L = \mathbb{E}Z_{2^L}$; the conditional error tends to zero for every $d > 0$, and bounded convergence proves $\mathbb{E}Y = \theta$. The bounds hold for every q with the same estimator and the same I_μ. Taking the two suprema separately and using $\operatorname{Var} Y \le \mathbb{E}Y^2$ proves theorem 13 and the improved upper constant in theorem 2. No interchange of a supremum with a strict inequality is needed.

Formal scope. `full_bound_713` proves the untruncated estimator result in a conditional observation-kernel model, including bounded independent observations with means $1/2 \pm D$. `uniform_family_bounds` supplies the separate-supremum conclusion for a fixed kernel and estimator over a nonempty family of latent laws sharing I_μ. These results do not yet realize the finite-budget calibration wrapper as a policy in the full fresh-coin experiment.

8 Consequences for power and logarithmic margins

Let $F(t) = \mu((0, t])$. Stieltjes integration by parts gives, for $0 < h < a$,

$$\omega_\mu(h) = F(h) + h\int_{(h,a]} \frac{1}{d}\,\mu(\mathrm{d}d) = \frac{h}{a} + h\int_h^a \frac{F(t)}{t^2}\,\mathrm{d}t. \tag{48}$$

The formula is valid with atoms. In particular,

$$I_\mu = \frac{1}{a} + \int_0^a \frac{F(t)}{t^2}\,\mathrm{d}t. \tag{49}$$

This integral, not a power exponent alone, determines exact unbiased feasibility.

8.1 The polynomial law as a corollary

For $F(t) = (t/a)^\alpha$ on $(0, a]$ and $u = h/a < 1$,

$$\omega_\mu(h) = \begin{cases} (\alpha u - u^\alpha)/(\alpha - 1), & \alpha \neq 1, \\ u(1 - \log u), & \alpha = 1. \end{cases} \tag{50}$$

Thus the main theorem immediately recovers the three regimes. The usual polynomial-margin class allows a more general constant:

$$\mathcal{G}_{\alpha,C} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \le t) \le Ct^\alpha, \quad 0 < t \le 1/2\}, \qquad C \ge 2^\alpha. \tag{51}$$

The last condition is necessary for nonemptiness.

Corollary 14 (Sharp polynomial-margin complexity)**.** *Fix $\alpha > 0$ and $C \ge 2^\alpha$. For all sufficiently small $\varepsilon > 0$ and $0 < \delta \le 1/4$, the smallest uniform expected work for an (ε, δ)-valid estimator on $\mathcal{G}_{\alpha,C}$ is*

$$\asymp_{\alpha,C} \log(1/\delta) \begin{cases} \varepsilon^{-2/\alpha}, & 0 < \alpha < 1, \\ \varepsilon^{-2}\log^2(1/\varepsilon), & \alpha = 1, \\ \varepsilon^{-2}, & \alpha > 1. \end{cases} \tag{52}$$

There is an unbiased estimator with uniformly finite variance and expected work on this entire class if and only if $\alpha > 1$. One common estimator works for every such α, C, without knowing either parameter.

Proof. Let $\mu_{\alpha,C}$ have CDF $\min\{Ct^\alpha, 1\}$ on $(0, 1/2]$. It is supported on $(0, C^{-1/\alpha}]$, and $\mathcal{G}_{\alpha,C} = \mathcal{G}_{\preceq\mu_{\alpha,C}}$ exactly. Its modulus is (50) with a replaced by $C^{-1/\alpha}$. Apply theorem 1 and corollary 3 and invert the three orders, with constants depending on (α, C). The inverse moment is finite exactly when $\alpha > 1$, so theorem 2 gives both directions of the unbiased dichotomy. Every polynomial envelope also has a finite uniform logarithmic moment; thus the universal estimator is integrable and unbiased even for $\alpha \le 1$, when finite uniform variance and work cannot both hold. □

8.2 Fixed-gap populations and rare-event scope

Corollary 15 (Gap classes)**.** *For $0 < \Delta \le 1/2$, put $\mathcal{G}_\Delta = \{G : \mathbb{P}_G(|P - 1/2| < \Delta) = 0\}$. For sufficiently small ε and $0 < \delta \le 1/4$, its minimax uniform expected work for (ε, δ) accuracy is*

$$\asymp \frac{\log(1/\delta)}{\varepsilon^2\Delta^2},$$

with universal comparison constants and a universal small-accuracy cutoff.

Proof. Here $\mathcal{G}_\Delta = \mathcal{G}_{\preceq\delta_\Delta}$ and $\omega_{\delta_\Delta}(h) = \min\{1, h/\Delta\}$. The envelope confidence law gives the result by direct inversion, taking ε smaller than its universal lower comparison constant. □

This is the balanced-population specialization of Lee–Valiant [1]. Their finer dependence on the positive fraction ρ is important for rare events; it is outside our worst case over all q. Our hard pairs keep every sign prior within $1/8$ of $1/2$, so both targets are near $1/2$. The localized theory is developed in section 11; a sharp characterization for every magnitude law remains open.

8.3 New boundary regimes invisible to a single exponent

For $\beta \geq 0$, define an actual CDF on $(0, a]$ by

$$F_\beta(t) = \frac{t/a}{[1 + \log(a/t)]^\beta}. \tag{53}$$

It is increasing, vanishes at zero, and equals one at a. With $L_h = 1 + \log(a/h)$, its modulus is exactly

$$\omega_\beta(h) = \frac{h}{a} \begin{cases} 1 + \dfrac{L_h^{1-\beta} - 1}{1-\beta}, & \beta \neq 1, \\ 1 + \log L_h, & \beta = 1, \end{cases} \qquad 0 < h < a. \tag{54}$$

This follows from (48) by the substitution $u = 1 + \log(a/t)$.

Corollary 16 (Logarithmically refined critical law)**.** *For the fixed magnitude law* (53)*, or its stochastic-dominance envelope class, the optimal expected work at small accuracy ε is, up to constants depending on β, a,*

$$\log(1/\delta) \times \begin{cases} \varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}, & 0 \leq \beta < 1, \\ \varepsilon^{-2}[\log\log(1/\varepsilon)]^2, & \beta = 1, \\ \varepsilon^{-2}, & \beta > 1. \end{cases} \tag{55}$$

Uniformly finite variance and work for unbiased estimation are possible exactly when $\beta > 1$, in which case $\mathbb{E}D^{-1} = \beta/[a(\beta - 1)]$.

Proof. Equation (54) and theorem 1 and corollary 3 give the error law. Its monotone inverse has the displayed orders: if $\omega(h) \asymp h[\log(1/h)]^\gamma$ with fixed $\gamma \geq 0$, then $h \asymp \varepsilon/[\log(1/\varepsilon)]^\gamma$; replacing the log by $\log\log$ gives the boundary case. Finally,

$$\mathbb{E}D^{-1} = \frac{1}{a}\left(1 + \int_1^\infty u^{-\beta}\,\mathrm{d}u\right),$$

which is finite exactly when $\beta > 1$. Apply theorem 2. □

For $\beta > 1$, this class permits efficient unbiased estimation even though it satisfies no bound $F(t) \leq Ct^\alpha$ with any $\alpha > 1$ and finite C. Indeed, $t^{1-\alpha}/[1 + \log(a/t)]^\beta \to \infty$. The earlier exponent boundary is therefore a corollary about entire polynomial classes, not the general structural criterion.

An iterated-log example. The same calculation extends further. Set $L_1(t) = 1 + \log(a/t)$, $L_2(t) = 1 + \log L_1(t)$, and $F(t) = (t/a)/(L_1(t)L_2(t)^\gamma)$ with $\gamma \geq 0$. Then

$$\omega(h) = \frac{h}{a}\left(1 + \int_1^{L_2(h)} v^{-\gamma}\,\mathrm{d}v\right).$$

At $\gamma = 1$, the fixed-confidence work is $\asymp \varepsilon^{-2}[\log\log\log(1/\varepsilon)]^2$, for sufficiently small ε. Unbiased feasibility again changes at $\gamma > 1$. This is an immediate instance of the general integral criterion, not an additional assumption or an independently designed algorithm.

Magnitude law near zero	Optimal work at fixed confidence	Efficient unbiased estimator
$F(t) \asymp t^\alpha$, $\alpha < 1$	$\varepsilon^{-2/\alpha}$	No
$F(t) \asymp t$	$\varepsilon^{-2}\log^2(1/\varepsilon)$	No
$F(t) \asymp t^\alpha$, $\alpha > 1$	ε^{-2}	Yes
$F(t) \asymp t/[\log(1/t)]^\beta$, $\beta < 1$	$\varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}$	No
$F(t) \asymp t/\log(1/t)$	$\varepsilon^{-2}[\log\log(1/\varepsilon)]^2$	No
$F(t) \asymp t/[\log(1/t)]^\beta$, $\beta > 1$	ε^{-2}	Yes

Table 1: Consequences of the modulus law. "Efficient unbiased" means both variance and expected work bounded uniformly over the corresponding sign or envelope class, not a claim for every individual population. For high confidence, multiply work by $\log(1/\delta)$.

9 Persistence under fixed finite-order density smoothness

The arbitrary-sign model makes no smoothness assumption on G. We now show that the critical logarithmic penalty is not solely caused by a jump in the density at the threshold.

For an integer $s \geq 1$, write $\|f\|_{C^s} = \max_{0\leq k\leq s} \|f^{(k)}\|_\infty$.

Theorem 17 (Smooth-density robustness)**.** *For each fixed integer $s \geq 1$, there exists a finite M_s such that, for every fixed $M \geq M_s$, the class of C^∞ probability densities supported in $[1/4, 3/4]$ and satisfying $\|f\|_{C^s} \leq M$ has minimax expected complexity*

$$\asymp_{s,M} \varepsilon^{-2}\log^2(1/\varepsilon)\log(1/\delta)$$

for sufficiently small ε and $0 < \delta \leq 1/4$. No unbiased estimator has both uniformly finite variance and uniformly finite expected work on this class.

Proof. Fix the local cutoff $a_0 = 1/4$. Choose a fixed even C^∞ density g supported strictly inside $(-a_0, a_0)$, with $g \geq c_g > 0$ on $[-a_0/2, a_0/2]$. The centered coordinate is $x = p - 1/2$. For sufficiently small $h > 0$, set

$$t_h = h^{1/(s+1)}, \qquad f_\pm(1/2 + x) = g(x)\left(1 \pm \frac{2hx}{x^2 + t_h^2}\right). \tag{56}$$

The perturbation is odd, so both densities integrate to one. Its absolute size is at most $h/t_h = h^{s/(s+1)}$, which is less than $1/4$ for small h, so the densities are nonnegative. The common magnitude density is $2g(d)$ and the conditional sign perturbation is

$$\eta_h(d) = \frac{hd}{d^2 + t_h^2}, \qquad \eta_h(d) \leq \frac{h}{2t_h} \leq \frac{1}{8}, \qquad d\eta_h(d) \leq h.$$

The posterior lemma therefore gives predictive KL at most $(8192/225)h^2$ per observation, uniformly over histories and selected coins. The target separation satisfies

$$\begin{aligned}\theta_+ - \theta_- &= 4h\int_0^{a_0} g(d)\frac{d}{d^2 + t_h^2}\,\mathrm{d}d \\ &\geq 2c_g h \log\frac{(a_0/2)^2 + t_h^2}{t_h^2} \geq c_s h\log(1/h)\end{aligned}$$

for all sufficiently small h, with $c_s > 0$; for a fixed bump it can be chosen proportional to $(s+1)^{-1}$. This is not uniform as $s \to \infty$.

For $R_t(x) = x/(x^2 + t^2) = t^{-1}R_1(x/t)$,

$$\|hR_{t_h}^{(k)}\|_\infty \leq C_k h t_h^{-(k+1)} = C_k h^{(s-k)/(s+1)} \leq C_k, \qquad 0 \leq k \leq s.$$

Leibniz's rule bounds every $\|f_\pm\|_{C^s}$ by a fixed M_s independent of h. They are individually C^∞ because $t_h > 0$ and g is a smooth compactly supported function. Uniform bounds on derivatives of all orders at once are not asserted.

Choose h of order $\varepsilon/\log(1/\varepsilon)$ with a sufficiently large fixed factor so that the separation exceeds 2ε. The stopped-transcript testing argument proves the lower bound. For the upper bound, $\|f\|_\infty \le M$ implies $\mathbb{P}(|P - 1/2| \le t) \le 2Mt$, so corollary 14 applies. The unbiased impossibility follows by averaging a putative uniform finite-variance, finite-work estimator and contradicting the lower bound at fixed confidence as $\varepsilon \downarrow 0$. □

Which smoothness is being controlled. This theorem concerns the density of the latent P. It is not a statement about smoothness of a regression function $y \mapsto \mathbb{E}[X \mid Y = y]$ in observable covariates. Smooth cross-state structure can enable information sharing, as in kernel methods [7]. Neither this theorem nor the next one rules out such improvements.

The construction does not settle analogous bounds for fixed analytic or Gevrey regularity classes; those require separate assumptions and arguments.

10 An observed-outer-state embedding

A latent coin and an observable outer state are different experiments. The following construction supplies a lower bound in the latter without silently identifying the two.

Let Y be uniform on $[0, 1]$. The algorithm may observe independent outer states and may also select any state at which to request a fresh inner observation. Let $X \in \{-1, 1\}$ and put $\theta = \mathbb{P}(\mathbb{E}[X \mid Y] > 0)$. There is no regularity assumption connecting the conditional means at different outer states.

Proposition 18 (Finite-cell observable-state lower bound)**.** *Fix a probability law μ on $(0, 1/2]$, $0 < h \le 1/2$, and $0 < \delta \le 1/8$. There is a finite partition of $[0, 1]$ and a family of deterministic, cellwise-constant conditional Bernoulli models such that: (a) their magnitude laws obey the envelope F_μ; (b) their construction has a scale separation $\Delta \ge \omega_\mu(h)/8$; and (c) every estimator valid to radius $\Delta/4$ and failure probability δ uniformly over this family has worst-case expected work at least*

$$\frac{225}{128h^2}\,\mathrm{kl}(1 - 2\delta, 2\delta). \tag{57}$$

This lower bound permits arbitrary selected-state queries and revisiting.

Proof. Round each $d \le h$ up to h. Round larger d upward to the next value in the grid $h, 2h, 4h, \ldots$, capped at a. This gives a finitely supported law μ_h on $\{d_1, \ldots, d_J\}$, with weights w_j, which is stochastically no smaller than μ. Above h the rounding increases a magnitude by at most a factor two; below h the clipped integrand stays one. Hence

$$\omega_{\mu_h}(h) \ge \tfrac{1}{2}\omega_\mu(h).$$

For each j, create M cells of lengths w_j/M and assign unknown fixed signs $s_{jk} \in \{0, 1\}$. On cell (j, k) let

$$B \mid Y \sim \mathrm{Ber}(1/2 + (2s_{jk} - 1)d_j), \qquad X = 2B - 1.$$

Every deterministic sign array has magnitude law μ_h, meets the envelope, and has target $\theta(s) = \sum_j (w_j/M) \sum_k s_{jk}$.

For the testing proof only, draw signs independently under two product priors with probabilities $1/2 \pm \eta_h(d_j)$ from (16). Their target means are separated by

$$\Delta = \tfrac{1}{4}\omega_{\mu_h}(h) \geq \tfrac{1}{8}\omega_\mu(h).$$

The sum of squared target weights is $\sum_j w_j^2/M \leq 1/M$. Weighted Hoeffding therefore gives

$$\mathbb{P}_\pm(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta/8) \leq 2\exp(-M\Delta^2/32).$$

Choose an integer $M \geq 32\Delta^{-2}\log(2/\delta)$. Uniform estimation error at most $\Delta/4$, together with the displayed concentration event, makes thresholding at $1/2$ a test with each error at most 2δ.

The signs are independent under either prior, so the local posterior lemma applies separately to every cell, regardless of the adaptively selected query points. Each inner observation has predictive KL at most $128h^2/225$. A uniform expected-work bound for deterministic sign arrays also bounds the prior-averaged expected work. Apply the KL chain rule and binary data processing to obtain (57). □

The partition size depends on the desired scale and confidence. This is a minimax construction in a class allowing arbitrarily fine finite partitions, not an asymptotic lower bound for one fixed finite list of cells. Since the outer distribution itself stays uniform on $[0,1]$, it establishes the relevant obstruction even with observable covariates, provided no cross-state regularity is imposed. The original upper algorithm can ignore the labels and use fresh outer states, so the envelope modulus remains the matching error scale for this unstructured class.

10.1 A continuous-magnitude subclass satisfying the nested assumptions

Proposition 19 (Continuous-magnitude observed-state obstruction)**.** *Fix the local cutoff $a_0 = 1/4$. Consider observed outer states $U \sim \mathrm{Unif}[0,1]$ and binary inner observations $X \in \{-1,1\}$ with*

$$|\mathbb{E}[X \mid U]| = 2D(U), \qquad D(U) \sim \mathrm{Unif}(0, a_0].$$

No cross-state regularity is assumed. There is a class of deterministic conditional models of this form whose minimax expected observation cost for error ε with fixed failure probability $1/16$ is

$$\asymp \varepsilon^{-2}\log^2(1/\varepsilon).$$

The lower bound allows arbitrary selected-state queries and revisiting. All models satisfy the bounded-density and normalized conditional-moment assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with constants independent of the error scale.

Proof. Fix $0 < h < a_0$ and partition $(0, a_0]$ into the first band $(0, h]$ and consecutive dyadic bands with right endpoints $2h, 4h, \ldots$, clipping the final endpoint at a_0. Let band j have endpoints ℓ_j, b_j and mass $w_j = (b_j - \ell_j)/a_0$. Partition $[0,1]$ into M equal-weight cells per band, each of length w_j/M. Within every cell in band j, let $D(U)$ increase linearly across $(\ell_j, b_j]$. Define its values at the finitely many endpoints to be positive and in the assigned band. Consequently the magnitude law is exactly uniform on $(0, a_0]$, not a discretization. The partition and the map D may be disclosed to the algorithm.

Assign each cell an unknown deterministic sign $s_{jk} \in \{0,1\}$. On that cell let $X = 2B - 1$ for $B \sim \mathrm{Ber}(1/2 + (2s_{jk} - 1)D(U))$. The target is

$$\theta(s) = \sum_j \frac{w_j}{M} \sum_{k=1}^{M} s_{jk}.$$

For the proof only, choose independent cell signs under two priors with probabilities $1/2 \pm \eta_j$, where $\eta_j = \frac{1}{8}\min\{1, h/b_j\}$. Their target means have separation

$$\Delta_h = 2\sum_j w_j \eta_j \geq \frac{1}{8}\omega_\mu(h), \qquad \mu = \mathrm{Unif}(0, a_0].$$

Indeed, within the first band the clipped integrand is one; within every other band $b_j \leq 2D(U)$. Every queried magnitude in band j satisfies $D(U)\eta_j \leq h/8$.

Repeated queries in a cell may use different magnitudes. Nevertheless, its sign has a single common likelihood ratio under the two priors, obtained by multiplying its queried sign-conditional likelihoods. The posterior identity and the predictive bound therefore still apply. Each observation contributes at most $128h^2/225$ KL, even under arbitrary query selection.

Weighted Hoeffding gives

$$\mathbb{P}_\pm\big(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta_h/8\big) \leq 2e^{-M\Delta_h^2/32}.$$

For $M \geq 32\Delta_h^{-2}\log(2/\delta)$, an estimator with uniform radius $\Delta_h/4$ and failure probability $\delta \leq 1/8$ gives a test of these priors with each error at most 2δ. The stopped-transcript argument gives expected work at least

$$\frac{225}{128h^2}\,\mathrm{kl}(1-2\delta, 2\delta).$$

For the uniform magnitude law, $\omega_\mu(h) = (h/a_0)[1+\log(a_0/h)]$. Set h of order $\varepsilon/\log(1/\varepsilon)$ with a sufficiently large fixed constant to obtain the lower rate. The upper rate follows by ignoring state labels and using the preceding threshold estimator. The class permits arbitrarily many cells; it is not a fixed finite-partition asymptotic claim.

Finally, $\mathrm{Var}(X \mid U) = 1 - 4D(U)^2 \geq 3/4$. Their normalized magnitude is

$$Z = \frac{|\mathbb{E}[X \mid U]|}{\sqrt{\mathrm{Var}(X \mid U)}} = \frac{2D}{\sqrt{1-4D^2}}.$$

It has density $[2a_0(1+z^2)^{3/2}]^{-1}$ on its support, bounded by $1/(2a_0)$. Moreover, $|X - \mathbb{E}[X \mid U]| \leq 3/2$, so for every finite $q > 2$ its standardized conditional qth absolute moment is bounded by $3^{q/2}$. These verify the two stated assumptions uniformly. A uniform RMS guarantee ε gives error at most 4ε with failure probability at most $1/16$, by Markov's inequality. Applying the fixed-confidence lower bound at 4ε yields the same RMS rate. The RMS upper bound follows from theorem 1. This also proves corollary 4. □

Interpretation. This identifies a hard subclass satisfying the explicit density and moment assumptions behind the classical adaptive nested-simulation upper rate. It does not rule out improvements from smooth cross-state regression structure, a known parametric model, or restrictions to rare-event populations. A restriction on the density of the latent magnitude is different from a restriction on the function of observed covariates.

11 Rare events and the inverse-square boundary

The preceding minimax law takes a worst case over every sign function. For rare events, fix $0 < \rho \leq 1/4$ and restrict this class to

$$\mathcal{Q}_\rho(\mu) = \left\{ q : (0, 1/2] \to [0,1] \text{ measurable} : \int q\, d\mu \leq \rho \right\}.$$

The unknown target is still $\theta(q) = \int q\, d\mu$; ρ is a known upper bound. Define

$$R_{\mu,\rho}(B) = \inf_{\mathcal{A}:\, \sup_{q\in\mathcal{Q}_\rho(\mu)} \mathbb{E}_q T_{\mathcal{A}} \le B} \sup_{q\in\mathcal{Q}_\rho(\mu)} \left(\mathbb{E}_q(\widehat{\theta}_{\mathcal{A}} - \theta(q))^2\right)^{1/2}.$$

The observation model remains the ordinary iid fresh-coin experiment: individual magnitudes and signs are hidden; all creation, revisiting, interleaving, and stopping decisions use only the seed and observed flips. The benchmark may know μ, ρ, B. In particular, the global constraint does not impose $q(d) \le \rho$ pointwise, nor fix either conditional magnitude law given the sign. Returning $\rho/2$ costs nothing and has RMS error at most $\rho/2$.

11.1 A variational lower modulus and its exact optimizer

For $h > 0$ set

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{v \text{ measurable},\ 0\le v\le 1 \\ \int v\, d\mu \le \rho}} \int \min\{v(d), h\sqrt{v(d)}/d\}\, \mu(dd). \tag{58}$$

There is a universal $c > 0$ such that $R_{\mu,\rho}(B) \ge c\Psi_{\mu,\rho}(B^{-1/2})$. The corresponding confidence lower bound replaces $B^{-1/2}$ by $\sqrt{\log(1/\delta)/B}$. The proof in appendix A explicitly imports the single-coin rare-prior Hellinger bound of Lee–Valiant [1, Corollary 19]; its bounded-ratio hybrid argument is written out. These statistical lower bounds are paper proofs with that published external input.

The variational optimization itself is fully explicit. There is a unique $t \in (0, 1/2)$ with

$$\mathbb{E}_\mu \min\{1, t^2/D^2\} = \rho.$$

For every $h > 0$, $v_*(d) = \min\{1, t^2/d^2\}$ is an optimizer. For $h \ge t$ the value is ρ; for $h < t$ it is

$$F_\mu(h) + h\mathbb{E}[D^{-1};\, h < D \le t] + ht\mathbb{E}[D^{-2};\, D > t]. \tag{59}$$

Atoms and singular laws are allowed, with no inverse-moment assumption. The Lean theorem `arbitrary_law_optimizer` proves this functional statement. It does not supply an attaining statistical algorithm. In particular,

$$\Psi_{\delta_a,\rho}(h) = \min\{\rho, h\sqrt{\rho}/a\}, \qquad \rho\omega_\mu(h/\sqrt{\rho}) \le \Psi_{\mu,\rho}(h) \le \omega_\mu(h).$$

11.2 The power-margin regimes

Let $\mu_{\alpha,a}$ have distribution function $(d/a)^\alpha$ on $(0, a]$ and put $x = (a\sqrt{B})^{-1}$. The proved RMS orders are:

Magnitude law	RMS order, with saturation included
$0 < \alpha < 1$	$\min\{\rho, x^\alpha\}$
$\alpha = 1$	$\min\{\rho, x[1 + \log_+(\rho/x)]\}$
$1 < \alpha < 2$	$\min\{\rho, \rho^{\,1-1/\alpha} x\}$
$\alpha = 2$, fixed $a = 1/4$	$\min\{\rho, \sqrt{\rho L(1 + \log L)/B}\}$
$\alpha > 2$	$\min\{\rho, \sqrt{\rho}\, x\}$

Here $L = 1 + \log(1/\rho)$. For the rows below two, $0 < a \le 1/4$; for the row above two, $0 < a \le 1/2$. Constants may depend on the fixed exponent. The quadratic row concerns one normalized law, with universal constants; it does not claim a theorem uniform in α near two. All rows hold for $B \ge 1$ and $0 < \rho \le 1/4$ in their stated domains.

For $0 < \alpha < 2$, appendix A gives full proofs, including confidence radii after replacing x by $a^{-1}\sqrt{\log(1/\delta)/B}$. The lower bound restricts the balanced hard alternatives to a region of magnitude mass ρ and does not use the imported rare-prior theorem. The upper bound starts the randomized telescope at a large enough cap that its base false-positive probability is at most ρ, then uses the same two-success calibration as section 5. It knows (α, a, ρ). It also holds under the envelope $F_\mu(d) \le \min\{(d/a)^\alpha, 1\}$.

For $\alpha > 2$, appendix B supplies an asymmetric sequential classifier: its negative-coin error is at most $\rho\varepsilon$, while its positive-coin error is at most ε. A randomized telescope then yields a one-coin unbiased output with separate bounds $\sup_q \mathbb{E}Y^2 \le (153/56)\rho$ and $\sup_q \mathbb{E}T_Y \le w_\alpha/a^2$, where $w_\alpha < \infty$ is explicit there. Independent Poisson replication calibrates every real budget, including expected replication counts below one. The upper bound is uniform over the same envelope class and knows (α, a, ρ, B). The matching exact-power lower bound uses the published fixed-gap Lee–Valiant theorem through a simulation that keeps a coin's privately sampled magnitude fixed on every revisit. Constants diverge as $\alpha \downarrow 2$. No sharp confidence characterization above or at two is asserted in this manuscript.

11.3 The full critical theorem and the correction to Ψ

Theorem 20 (Critical quadratic law)**.** *Fix $\mu(dd) = 32d\,dd$ on $(0, 1/4]$. There are universal $0 < c < C < \infty$ such that for all $0 < \rho \le 1/4$ and $B \ge 1$,*

$$c\min\{\rho, \sqrt{\rho LH/B}\} \le R_{\mu,\rho}(B) \le C\min\{\rho, \sqrt{\rho LH/B}\}, \qquad L = 1+\log(1/\rho), \quad H = 1+\log L.$$

The lower bound covers every observable randomized population algorithm, including nonlinear outputs, unbounded stopping, and arbitrary creation and interleaving. The upper construction may know μ, ρ, B.

Evidence: complete paper argument in appendix C; its full statistical chain remains unformalized.

The proof has three substantive transitions. First, sequential testing at many possible gaps forces a scale-entropy cost. Conditional moments, activation normalization, and repeated runs on unused flips of the *same* hidden coin transfer that fact to a small-bias one-coin estimator: its variance supremum times its expected work under the all-negative population is at least $c_0\rho LH$. Second, a finite-experiment RMS comparison and compact convex minimax argument obtain *one* population pair for every capped single-coin policy. A policy-dependent hard pair would not suffice. Third, a finite-prefix midpoint decomposition transfers this pair to interacting population algorithms, charging each flip once. The same pair works at every cap; finite expected work under both endpoints justifies passage to terminal outputs. The longer proof, including the upper construction, is in appendix C.

For this fixed law, the optimizer identity is

$$\rho = (t/a)^2[1+2\log(a/t)], \qquad \Psi_{\mu,\rho}(h) = \begin{cases} [2ht(1+\log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \ge t, \end{cases} \qquad a = 1/4.$$

It follows that $\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L/B}\}$. Taking $\rho_n = \exp(-\exp n)$ and $B_n = \rho_n^{-2}$ gives

$$\frac{R_{\mu,\rho_n}(B_n)}{\Psi_{\mu,\rho_n}(B_n^{-1/2})} \ge c'\sqrt{1+\log(1+\log(1/\rho_n))} \longrightarrow \infty.$$

Thus the proposed universal matching-Ψ upper bound is false according to theorem 20, even for one known law and known rarity. The variational optimizer remains valid. A sharp replacement for arbitrary laws must account for additional statistical difficulty across scales; no such replacement is established here.

11.4 The formal entropy milestone

The part of the critical proof now verified in Lean is a statistical finite-gap theorem, rather than just its concluding scalar inequality. Let $0 < d_j \le 1/128$, $d_j/d_{j+1} \ge 2$, and $\log K \ge 32$. One measurable test with an independent seed and observable stopping time T is run on iid Bernoulli flips of mean $1/2 \pm d_j$. If it has finite expected work and error at most $1/10$ under both signs for every gap, then

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j} T \ge \frac{K \log K}{512}.$$

`finite_gap_sign_entropy` derives this from the actual infinite Bernoulli experiment. It assumes neither a KL bound nor a packing estimate nor termination under the fair coin. This proves the finite-gap ingredient in lemma 39. Its continuum restart application, the common population pair, and the population transfer remain paper arguments.

12 Verification, provenance, and remaining questions

The distinction between a checked theorem and a checked ingredient is part of each result's scope. The following table records the boundary at the finite-gap checkpoint `accdb144972e dd255dedec50cc1cbcaa45587040`. Links point to that source revision rather than a moving branch.

Claim	Verification scope
Original RMS minimax lower bound	`minimax_lower_bound` proves $R_\mu(B) \ge \omega_\mu(B^{-1/2})/64$ in the hidden-magnitude iid fresh-coin model, for arbitrary measurable seeds, natural-number coin labels, revisits, interleaving, and unbounded stopping. It constructs the experiment and retains infinite risk. The stronger revealed- or selected-magnitude paper statement is not the exact Lean target.
Original upper and confidence laws	The budget-only algorithm and confidence characterization in theorem 1 remain paper proofs. Classifier estimates and inverse-binomial identities alone do not establish their realization in the full algorithm model.
Unbiased characterization	The necessity of $I_\mu < \infty$ and lower product coefficient $1/16$ in theorem 2 remain paper proofs. The constructive upper bound has the formal coverage described in the next row.
713 construction	`full_bound_713` proves the untruncated estimator theorem in a conditional observation-kernel model. `uniform_family_bounds` proves the separate-supremum bridge with a fixed kernel and estimator.

Claim	Verification scope
Variational optimizer	`arbitrary_law_optimizer` proves the optimizer and value formula for arbitrary laws. It does not assert a matching rare-event statistical upper bound.
Rare-event rates	The rates in section 11 remain paper theorems. The arbitrary-law lower bound and the above-two matching lower bound use the cited Lee–Valiant input. The critical argument has the formal components in the next two rows.
Finite-gap sign entropy	`finite_gap_sign_entropy` proves the statistical bound with actual stopping and flip costs. The audit checks five exact targets and all 63 public declarations in its component chain.
Common-point step	`common_point_of_finite_mixtures` proves an abstract compact-convex theorem. Statistical compactness, mixture information, and the hypothesis needed to instantiate it are not supplied by that theorem.

The audits permit three standard Lean axioms: `propext`, `Classical.choice`, and `Quot.sound`. The exact specifications and model definitions are necessary to interpret that fact: absence of proof holes does not show that a statement matches an intended experiment. The finite-gap targets were introduced with their proofs and are not described as specifications frozen before the proof work. The repository's verification guide gives reproduction commands and the source-specific audit records.

Numerical evidence and its limits. Exact finite certificates, floating-point recursions, and held-out simulations do not certify the continuum theorems or the iterated-log separation. Finite-grid constants do not replace 713: the grid-winning logarithmic threshold coefficient 1/2 fails the uniform error envelope at an explicit large cap, whereas this construction retains coefficient one. The finite-budget feedback advantage and practical experiments remain supporting research with their original limited scopes.

Provenance and review. This version incorporates the original law, the 713 argument, and the rare-event continuations, with full proofs in the main text and appendices A to C. The assembly script checks source identities and records editorial changes; historical sources and reviews remain unchanged. AI agents performed the work under David Iach's direction. No external specialist review or model diversity is established. One critical reviewer authored part of the argument, precluding two independent validations of the full proof. The process note records the workflow, failed approaches, and these limits.

Remaining mathematical work. The critical continuum null-cost theorem, statistical common-pair construction, and interacting-population transfer are the immediate verification targets. The original distribution-blind upper theorem still needs one observable budget-indexed algorithm realized in the same full experiment as the formal lower bound. The unbiased converse and confidence characterization also remain unformalized. A corrected arbitrary-law rare-event functional is open.

The model requires independent fresh coins and conditionally independent flips, without drift or additional smooth structure linking states. Expected-work optimality supplies no deterministic latency bound or data-computable confidence radius over all unknown laws. Neither useful finite critical constants nor practical superiority is established.

A Rare-event localization and margins below two

This appendix supplies the general lower bound, its optimizer, the below-two matching construction, and a general upper bracket. The earlier arbitrary-law conjecture has been replaced by its correction. The published rare-prior input is identified explicitly below.

A.1 Experiment, localization, and the two anchors

A fresh coin has $D \sim \mu$ on $(0, 1/2]$, $S \mid D = d \sim \mathrm{Ber}(q(d))$, and probability

$$P = \tfrac{1}{2} + (2S - 1)D.$$

Conditional on its latent probability, successive observations of that coin are independent Bernoulli trials. Distinct fresh coins are independent. An algorithm can create coins, revisit them, interleave their observations, and stop adaptively. It observes neither D nor S. One observation is one unit of work. Arithmetic and independent randomization are not charged, as in the original model.

Fix $0 < \rho \le 1/4$ and define

$$\mathcal{Q}_\rho(\mu) = \left\{ q : (0, 1/2] \to [0, 1] \text{ measurable} : \quad \theta(q) := \int q \, d\mu \le \rho \right\}. \tag{60}$$

The quantity to estimate is $\theta(q)$, not the known upper bound ρ. Replacing the inequality by the known equality $\theta = \rho$ would trivialize estimation.

Let $R_{\mu,\rho}(B)$ be the infimum, over algorithms with $\sup_{q \in \mathcal{Q}_\rho} \mathbb{E}_q T \le B$, of $\sup_q (\mathbb{E}_q |\widehat{\theta} - \theta(q)|^2)^{1/2}$. Let $r_{\mu,\rho}(B, \delta)$ be the corresponding smallest uniform error radius at failure probability at most δ. Here $B \ge 1$ and $0 < \delta \le 1/4$. These are *expected-work* criteria. The minimax benchmark may know μ, ρ. Each construction's actual parameter requirements are stated separately.

This class deliberately permits the positives to concentrate in a difficult subpopulation. It does *not* impose the stronger condition $q(d) \le \rho$ at every magnitude. Nor does it assume known conditional magnitude laws μ_+, μ_-. In the latter model the mixture weight is the main unknown; in (60), changing q changes both conditional laws. Lee–Valiant's Section 7 addresses the known-conditional-law experiment with a per-coin depth restriction [1].

A.1.1 Sanity checks

For $\mu = \delta_\Delta$, the problem is the fixed-gap mixture experiment. The Lee–Valiant bounds [1], localized to $\theta \le \rho$, give, in the nontrivial regime $\varepsilon \le c\rho$,

$$B^*(\varepsilon, \delta) \asymp \frac{\rho}{\varepsilon^2 \Delta^2} \log(1/\delta). \tag{61}$$

The corresponding error scale, saturated at the trivial radius, is

$$\min\{\rho, \sqrt{\rho \log(1/\delta)/B}/\Delta\}.$$

At fixed confidence or for RMS, omit the logarithmic factor. This is a cited anchor, not a new fixed-gap theorem.

For constant ρ, a correct localized scale must agree, up to constants, with the original

$$\omega_\mu(h) = \mathbb{E}_\mu \min\{1, h/D\}, \qquad h = B^{-1/2}$$

(and $h = \sqrt{\log(1/\delta)/B}$ for high confidence). The proposed heuristic

$$H_{\mu,\rho}(h) = \rho \, \omega_\mu(h/\sqrt{\rho}) \tag{62}$$

satisfies both anchors. The next section shows that this is insufficient.

A.2 A counterexample to the proposed rescaling

Theorem 21. *Let* $0 < \rho \le 1/4$, $a = 1/4$, $d = \rho^2$,

$$\mu_\rho = \rho\delta_d + (1-\rho)\delta_a, \qquad B = \rho^{-4}.$$

Then $R_{\mu_\rho,\rho}(B) \ge \rho/100$, *whereas*

$$H_{\mu_\rho,\rho}(B^{-1/2}) = \rho^2 + 4(1-\rho)\rho^{5/2} = O(\rho^2).$$

The minimax error is therefore of order ρ, *and the heuristic can underestimate it by an unbounded factor.*

Proof. Put $\eta = 1/64$ and choose $q_\pm(d) = 1/2 \pm \eta$, $q_\pm(a) = 0$. Both targets lie in $[0,\rho]$ and are separated by $\Delta = 2\eta\rho = \rho/32$. Even if the algorithm is told every magnitude, only a queried coin of magnitude d is informative. The original posterior inequality gives, for every next observation and every history,

$$\mathrm{KL}_{\mathrm{next}} \le \frac{32d^2\eta^2}{(1-4\eta^2)^2}.$$

The stopped chain rule and $Bd^2 = 1$ imply $\mathrm{KL}(\mathsf{P}_+, \mathsf{P}_-) \le k_0 := 32\eta^2/(1-4\eta^2)^2 < 0.008$. Hence $\mathrm{TV} \le \sqrt{k_0/2}$. Thresholding an estimator halfway between the two targets gives error sum at most $8R^2/\Delta^2$, whereas every test has error sum at least $1 - \mathrm{TV}$. Thus

$$R \ge \Delta\sqrt{(1-\sqrt{k_0/2})/8} > \rho/100.$$

The constant estimator $\rho/2$ has error at most $\rho/2$. Finally, $(\rho B)^{-1/2} = \rho^{3/2}$ lies between d and a. Direct substitution in (62) gives the stated expression. □

The hard subpopulation contains both positive and negative near-threshold coins. Its local positive fraction is close to $1/2$, although its total population mass is only ρ. Multiplying the global modulus by ρ incorrectly treats the difficult portion as if positives were thinly spread across all magnitudes. Section A.4 also gives a failure example with one fixed, uniform magnitude law, so this is not merely an artifact of changing μ with ρ.

A.3 Two localized lower bounds for arbitrary laws

A.3.1 A balanced-region bound independent of rare-prior machinery

Define

$$\Gamma_{\mu,\rho}(h) = \sup_{A:\,\mu(A)\le\rho} \int_A \min\{1, h/d\}\,\mu(dd). \tag{63}$$

Proposition 22. *There is a universal* $c > 0$ *such that*

$$R_{\mu,\rho}(B) \ge c\Gamma_{\mu,\rho}(B^{-1/2}), \qquad r_{\mu,\rho}(B,\delta) \ge c\Gamma_{\mu,\rho}\left(\sqrt{\log(1/\delta)/B}\right).$$

This lower bound remains valid with revealed or selected magnitudes.

Proof. For any admissible A, put $q_\pm(d) = \mathbf{1}_A(d)[1/2 \pm \eta_h(d)]$ with $\eta_h(d) = \frac{1}{8}\min(1, h/d)$. The larger target is at most $5\mu(A)/8 \le \rho$. The target separation is $\Delta_h = \frac{1}{4}\int_A \min(1, h/d)\,d\mu$.

Every observation has predictive KL at most $128h^2/225$ by section 3, and observations outside A have zero divergence. Thus transcript KL is at most $128Bh^2/225$. For RMS choose $h = c_0 B^{-1/2}$ with a sufficiently small universal c_0 and use the testing argument above. For confidence use $h = c_1\sqrt{\log(1/\delta)/B}$, choosing c_1 so that transcript KL is strictly less than $\mathrm{kl}(1-\delta, \delta)$; an estimator with radius smaller than $\Delta_h/2$ would contradict data processing. The integrand satisfies $\min(1, \lambda h/d) \geq \lambda \min(1, h/d)$ for $0 < \lambda \leq 1$. Taking the supremum over A proves the claims without changing the scale. □

Atoms can prevent Γ from using a small amount of mass at a magnitude. The next functional permits fractional allocation and includes the fixed-gap rarity factor.

A.3.2 A variational lower modulus

Define

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0 \leq w \leq 1 \\ \int w\, d\mu \leq \rho}} \int \min\{w(d),\, h\sqrt{w(d)}/d\}\, \mu(dd). \tag{64}$$

The optimization variable w represents how a hard alternative allocates its available positive mass. It is not an assumed or estimated sign function of the true population.

Lemma 23 (Published single-coin input)**.** *There are universal constants $c_*, C_* > 0$ such that, for $0 < u \leq c_*$, $0 \leq v \leq c_* u$, $0 < d \leq 1/2$, and any stopped single-coin adaptive experiment, its transcript laws under sign priors u and $u + v$ satisfy*

$$H^2(\mathsf{P}_u, \mathsf{P}_{u+v}) \leq C_* \frac{v^2 d^2}{u} \mathbb{E}_u \tau. \tag{65}$$

Here $H^2(P, Q) = 1 - \int \sqrt{dP\, dQ}$ and external, parameter-independent randomization is allowed.

Source and extension. For sufficiently small d, this is the first expected-work bound in Lee–Valiant Corollary 19 [1], with their ρ, ϵ, Δ replaced by u, v, d. Their result averages over histories and is not a history-uniform predictive-KL claim. Its proof is not reproduced here; it is an explicit external theorem used by this note. For $d \geq d_* > 0$, reveal the sign at the first requested observation. Before that request the experiment has no sign information. By data processing, the Hellinger distance is at most the probability of making the first request times the distance between $\mathrm{Ber}(u)$ and $\mathrm{Ber}(u + v)$. The latter is at most Cv^2/u when u and v/u are small. Since $\mathbb{E}_u \tau$ is at least that request probability and $d^2 \geq d_*^2$, this proves (65) on the remaining range, including deterministic coins at $d = 1/2$. Increasing a universal constant joins the two ranges. □

Lemma 24 (Bounded-ratio decomposition)**.** *Consider two independent-coin population laws ν_0, ν_1 whose latent Radon–Nikodym ratio lies in $[1-\kappa, 1+\kappa]$ for some fixed $\kappa < 1$. For a fully adaptive experiment stopped after at most n observations, let P_0, P_1 be its transcript laws. Let P_i be the hybrid law using ν_1 only for coin i and ν_0 for every other coin. Then*

$$\mathrm{KL}(P_0, P_1) \leq C_\kappa \sum_i H^2(P_0, P_i).$$

The same conclusion holds by finite-prefix limits for almost-surely terminating experiments whenever the right-hand side is uniformly bounded along the prefixes.

Proof. Index coins in order of their first observation, so at most n indices occur. Include the algorithm's independent random seed in the transcript. For a fixed transcript, let $L_i =$

dP_i/dP_0. This is the local coin likelihood ratio, marginalized over that coin's latent state, so $1-\kappa \le L_i \le 1+\kappa$. Independence of latent coins and the parameter-independent action rule give

$$\frac{dP_1}{dP_0} = \prod_i L_i.$$

This identity holds even though the local sequence lengths were chosen using other coins' outcomes. Also $\mathbb{E}_0 L_i = 1$. On this compact ratio interval, $-\log x + (x-1) \le C_\kappa (x-1)^2$, and $(x-1)^2 \le C_\kappa(\sqrt{x}-1)^2$. Taking expectations and using $\mathbb{E}_0(\sqrt{L_i}-1)^2 = 2H^2(P_0,P_i)$ proves the finite-prefix claim. Increasing stopped-transcript sigma fields and the standard monotone limit of relative entropy prove the last assertion. Applying the argument at finite prefixes avoids any unjustified interchange of an unbounded stopping sum and a likelihood derivative. □

This lemma is the bounded-relative-change mechanism of Lee–Valiant Lemma 18, stated for the present magnitude-dependent perturbations. It is not a new general decomposition principle. Unlike an attempted uniform posterior bound, it charges informative but rare histories by their expected frequency.

Theorem 25 (Arbitrary-law rare-event lower bound)**.** *For the ordinary iid fresh-coin experiment, there is a universal $c > 0$ such that*

$$R_{\mu,\rho}(B) \ge c\Psi_{\mu,\rho}(B^{-1/2}), \tag{66}$$

$$r_{\mu,\rho}(B,\delta) \ge c\Psi_{\mu,\rho}\left(\sqrt{\log(1/\delta)/B}\right). \tag{67}$$

The proof uses Lemma 23; it does not assert this strengthened bound for an arbitrary selected-magnitude oracle.

Proof. Fix any feasible w in (64) and write $f_h(d) = \min\{w(d), h\sqrt{w(d)}/d\}$. Choose small universal constants $b > 0$ and $c_0 > 0$ with $c_0 \ll b$ and $b + c_0 \le 1$. Define

$$q_0(d) = bw(d), \qquad q_1(d) = bw(d) + c_0 f_h(d).$$

Both belong to (60). Their target separation is $c_0 \int f_h \, d\mu$. Where $w > 0$, put $u = bw$, $v = c_0 f_h$. The relative change v/u is uniformly small and

$$\frac{v^2 d^2}{u} \le \frac{c_0^2}{b} h^2. \tag{68}$$

Where $w = 0$ the two laws agree. The sign-prior likelihood ratios, and hence the latent coin ratios, are uniformly in $[1-\kappa, 1+\kappa]$ for a fixed $\kappa < 1$.

At a finite observation prefix, apply Lemma 24. To bound its ith term, simulate all other coins under q_0 as external randomness, and reveal D_i only for the purpose of bounding information. Conditional on $D_i = d$, this is a single-coin stopped experiment with priors $u(d), u(d)+v(d)$. Data processing, Lemma 23, and (68) give

$$H^2(P_0,P_i) \le Ch^2 \mathbb{E}_0 \tau_i.$$

Here the conditional distributions of all simulated coins and the algorithmic seed are the same in both worlds. Summing over coins counts each observation once. Thus

$$\mathrm{KL}(P_0,P_1) \le Ch^2 \mathbb{E}_0(T \wedge n) \le Ch^2 B.$$

Let n tend to infinity. For RMS choose a small constant multiple of $B^{-1/2}$; for confidence choose a small constant multiple of $\sqrt{\log(1/\delta)/B}$. The same two-point testing arguments as Proposition 22 apply. Finally, $\Psi(\lambda h) \ge \lambda \Psi(h)$ for $0 < \lambda \le 1$, and taking the supremum over w yields the stated scales. □

A.3.3 An exact solution of the variational problem

Proposition 26. *For $0 < \rho < 1$, let $t = t_{\mu,\rho} > 0$ be the unique solution of*

$$\mathbb{E}_\mu \min\{1, t^2/D^2\} = \rho. \tag{69}$$

Then $w_*(d) = \min\{1, t^2/d^2\}$ is an optimizer in (64) for every $h > 0$. If $h \geq t$, then $\Psi_{\mu,\rho}(h) = \rho$. If $0 < h < t$, then

$$\Psi_{\mu,\rho}(h) = F_\mu(h) + h\mathbb{E}_\mu[D^{-1};\, h < D \leq t] + ht\mathbb{E}_\mu[D^{-2};\, D > t]. \tag{70}$$

These identities allow atoms and singular laws.

Proof. The left side of (69) is continuous, tends to zero at $t = 0$, and is strictly increasing until it reaches one. This proves existence and uniqueness at $\rho < 1$. For $h < t$, put $\lambda = h/(2t) < 1/2$. For fixed d, the concave function $f_d(w) = \min\{w, (h/d)\sqrt{w}\}$ is linear until $w = (h/d)^2$, then square-root, on $[0, 1]$. Maximizing $f_d(w) - \lambda w$ on $[0, 1]$ gives exactly $w_*(d)$: in the square-root portion its stationary point is t^2/d^2, and otherwise the maximum is at 1. Integration and $\int w_*\, d\mu = \rho$ prove optimality. Splitting at h and t gives (70), with the endpoint contributions agreeing if there are atoms. If $h \geq t$, $f_d(w_*) = w_*$ everywhere, so the objective equals the maximum possible value ρ. □

In particular,

$$\Psi_{\delta_\Delta,\rho}(h) = \min\{\rho, h\sqrt{\rho}/\Delta\}, \qquad \rho\omega_\mu(h/\sqrt{\rho}) \leq \Psi_{\mu,\rho}(h) \leq \omega_\mu(h). \tag{71}$$

The heuristic is the feasible *constant allocation* $w \equiv \rho$ in the variational problem. Its failure is an allocation failure, not a failure of its two sanity anchors. If ρ stays bounded away from zero, (71) recovers the original modulus up to constants. The inverse-square tail in (70) describes how hard negative coins outside the unresolved core still contribute.

Correction to the earlier conjecture. The proposed arbitrary-law matching-Ψ upper bound is refuted by Theorem 20. The lower bound and optimizer above remain valid. No matching replacement for arbitrary laws is supplied here.

A.4 Matching rare-event rates for power margins

Let $\mu_{\alpha,a}$ have $F(t) = (t/a)^\alpha$ on $(0, a]$, with $0 < a \leq 1/4$. Write

$$\mathcal{R}_{\alpha,\rho}(x) = \begin{cases} \min\{\rho, x^\alpha\}, & 0 < \alpha < 1, \\ \min\{\rho, x[1 + \log_+(\rho/x)]\}, & \alpha = 1, \\ \min\{\rho, \rho^{1-1/\alpha}x\}, & 1 < \alpha < 2. \end{cases} \tag{72}$$

Theorem 27. *For fixed $0 < \alpha < 2$, all $0 < a \leq 1/4$, $0 < \rho \leq 1/4$, $B \geq 1$, and $0 < \delta \leq 1/4$,*

$$R_{\mu_{\alpha,a},\rho}(B) \asymp_\alpha \mathcal{R}_{\alpha,\rho}((a\sqrt{B})^{-1}),$$
$$r_{\mu_{\alpha,a},\rho}(B, \delta) \asymp_\alpha \mathcal{R}_{\alpha,\rho}(a^{-1}\sqrt{\log(1/\delta)/B}).$$

The same upper bounds hold for every magnitude law with $F_\mu(t) \leq \min\{(t/a)^\alpha, 1\}$. The upper construction knows (α, a, ρ), not the individual magnitudes or sign function. It does not give a distribution-blind rare-event algorithm for arbitrary laws.

A.4.1 Lower proof

Take $A = (0, a\rho^{1/\alpha}]$ in Proposition 22. Its mass is ρ. Put $t_\rho = a\rho^{1/\alpha}$. For $h \geq t_\rho$ the restricted modulus is ρ. For $h < t_\rho$ it is

$$\left(\frac{h}{a}\right)^\alpha + \frac{\alpha h}{a^\alpha} \int_h^{t_\rho} d^{\alpha-2}\, dd. \tag{73}$$

For $\alpha \neq 1$, the integral is $(t_\rho^{\alpha-1} - h^{\alpha-1})/(\alpha - 1)$; for $\alpha = 1$ the result is $(h/a)[1 + \log(a\rho/h)]$. These expressions are comparable, with constants depending only on α, to (72). This proof uses the balanced rare-region construction and does not need the imported single-coin rare-prior lemma.

A.4.2 Why the original unmodified variance is not rarity-scaled

Proposition 28. *Fix an interior magnitude $0 < d < 1/2$ and let every coin be negative. For the original untruncated geometric single-term estimator Y, which is unbiased and square-integrable at this fixed gap,*

$$\operatorname{Var} Y \geq \frac{1/2 - d}{1 - 2^{-1/2}} > 0.$$

For the truncation at level L, the same second-moment lower bound holds, and $\operatorname{Var} Y_L \geq (1/2 - d)/(1 - 2^{-1/2}) - (\mathbb{E} Z_{2^L})^2$.

Proof. The event $J = 0$ has probability $p_0 = 1 - 2^{-1/2}$ and contributes $Y = X/p_0$ for one Bernoulli observation with success probability $1/2 - d$. Therefore $\mathbb{E}Y^2 \geq (1/2 - d)/p_0$. In the untruncated case $\mathbb{E}Y = 0$. In the truncated case telescoping gives $\mathbb{E}Y_L = \mathbb{E}Z_{2^L}$; subtract its square. □

This is an analytic obstruction to deriving a ρ factor by reusing the original second-moment estimate unchanged. At $d = 1/4$ the lower floor is approximately 0.8536, even though $\theta = 0$.

A.4.3 A shifted-start construction

Retain the original classifier Z_n with dyadic checkpoints and threshold $\sqrt{[1 + \log(n/k)]/k}$ at checkpoint $k < n$. Define

$$e_n(d) = \min\{1, (nd^2)^{-2}\}, \qquad m_n(d) = \min\{n, d^{-2}[1 + \log_+(nd^2)]\}.$$

The original conditional inequalities are $\mathbb{P}(Z_n \neq S \mid d, S) \leq e_n(d)$ and $\mathbb{E}(\tau_n \mid d, S) \leq 17 m_n(d)$ (sections 4 and 5).

Lemma 29. *For $na^2 \geq 1$ and $0 < \alpha < 2$,*

$$e_n := \mathbb{E}_{\mu_{\alpha,a}} e_n(D) \leq K_e a^{-\alpha} n^{-\alpha/2}, \qquad K_e = \frac{4}{4 - \alpha}, \tag{74}$$

$$\mathbb{E}_{\mu_{\alpha,a}} m_n(D) \leq K_m a^{-\alpha} n^{1-\alpha/2}, \qquad K_m = 1 + \frac{\alpha}{2 - \alpha} + \frac{2\alpha}{(2 - \alpha)^2}. \tag{75}$$

The same bounds hold under the stated stochastic-dominance envelope.

Proof. Split each integral at $d = n^{-1/2}$. For (74), the first part is $(a\sqrt{n})^{-\alpha}$; extending the remaining power integral to infinity gives the factor $1 + \alpha/(4 - \alpha)$. For (75), substitute $u = d\sqrt{n}$ in the tail and extend its upper endpoint to infinity. The remaining factor is

$$1 + \alpha \int_1^\infty u^{\alpha-3}(1 + 2\log u)\, du = K_m.$$

Both integrands are nonincreasing in d, which proves the envelope statement. The restriction $\alpha < 2$ is essential to this particular work bound. □

Choose the smallest dyadic n_0 satisfying

$$n_0 \geq a^{-2}(K_e/\rho)^{2/\alpha}. \tag{76}$$

If $B < 4n_0$, return $\rho/2$. Otherwise choose $N = n_0 2^L$ as the largest such number with $4N \leq B$. Thus $B/8 < N \leq B/4$. Set $n_j = n_0 2^j$ and

$$\Delta_0 = Z_{n_0}, \qquad \Delta_j = Z_{n_j} - Z_{n_{j-1}}, \quad j \geq 1,$$

using a shared observation prefix within each increment. Sample J with $p_j = (1-r)r^j$, $r = 2^{-1/2}$, and return $Y_L = \Delta_J/p_J$ if $J \leq L$ and zero otherwise. Coins are independent between calls. The observed work C_L is at most N.

Lemma 30. *With $S_L = \sum_{j=0}^{L} n_j^{(1-\alpha)/2}$, uniformly over the rare-event envelope class,*

$$|\mathbb{E}Y_L - \theta| \leq K_e a^{-\alpha} N^{-\alpha/2}, \tag{77}$$

$$\mathbb{E}Y_L^2 \leq C_\alpha a^{-\alpha} n_0^{-1/2} S_L, \tag{78}$$

$$\mathbb{E}C_L \leq C_\alpha a^{-\alpha} n_0^{1/2} S_L. \tag{79}$$

Proof. Telescoping proves (77). At the base, $\mathbb{E}Z_{n_0}^2 = \mathbb{P}(Z_{n_0} = 1) \leq \theta + e_{n_0} \leq 2\rho$. Dyadic rounding in (76) gives $\rho \leq 2^{\alpha/2} K_e a^{-\alpha} n_0^{-\alpha/2}$, so the base obeys the same power envelope as the increments. For $j \geq 1$,

$$\mathbb{E}\Delta_j^2 \leq e_{n_j} + e_{n_{j-1}} \leq C_\alpha a^{-\alpha} n_j^{-\alpha/2}.$$

Divide by $p_j = (1-r)\sqrt{n_0/n_j}$ and sum to obtain (78). For work, a coupled increment costs at most the sum of the two classifier costs, so its expectation is at most $34K_m a^{-\alpha} n_j^{1-\alpha/2}$; the base needs only the factor 17. Multiply by p_j and sum to obtain (79). □

A.4.4 Calibration and completion of the upper proof

Apply the original two-success wrapper to iid pairs (Y_L, C_L), with cap N. Its pilot markers have success probability C_L/N. Stop at the second success, at pilot count K, discard the pilot values, and average $K-1$ fresh production values. With $c = \mathbb{E}C_L > 0$ and $v = \operatorname{Var} Y_L$, the identities are

$$\mathbb{E}T = 4N - c \leq B, \quad \mathbb{E}\widehat{\theta} = \mathbb{E}Y_L, \quad \operatorname{Var}\widehat{\theta} = vc/N. \tag{80}$$

The identity $\mathbb{E}[1/(K-1)] = c/N$ is classical inverse-binomial estimation; the wrapper is inherited from section 5, not claimed as new [8, 16]. Independence of the fresh production samples is essential.

Equations (77)–(79) imply

$$\left(\mathbb{E}|\widehat{\theta} - \theta|^2\right)^{1/2} \leq C_\alpha \left[a^{-\alpha} N^{-\alpha/2} + a^{-\alpha} N^{-1/2} S_L\right]. \tag{81}$$

For $\alpha < 1$, $S_L \leq C_\alpha N^{(1-\alpha)/2}$. For $\alpha = 1$, $S_L = L + 1$. For $\alpha > 1$, $S_L \leq C_\alpha n_0^{(1-\alpha)/2}$. Substitution of (76) and $N \asymp B$ yields (72). In the small-budget branch, the right side of the target rate is at least a constant multiple of ρ, so returning $\rho/2$ suffices. Projecting the final estimate onto $[0, \rho]$ cannot increase squared error.

For high confidence, use an odd number $k \asymp \log(1/\delta)$ of independent groups, each with expected budget B/k, and take their median. Chebyshev gives failure at most $1/8$ at $\sqrt{8}$ times each group's RMS bound; a binomial tail bound gives median failure at most $e^{-9k/32}$. The rate follows after replacing B by B/k. If group budgets are too small, the constant estimator handles the saturated regime. Expected costs add. This completes Theorem 27.

A.4.5 Consequences and limits of the theorem

For $0 < \varepsilon \le c_\alpha \rho$, the required expected work is, up to α-dependent factors,

Margin	Work, multiplied by $a^{-2}\log(1/\delta)$
$0 < \alpha < 1$	$\varepsilon^{-2/\alpha}$
$\alpha = 1$	$\varepsilon^{-2}[1 + \log(\rho/\varepsilon)]^2$
$1 < \alpha < 2$	$\rho^{2-2/\alpha}\varepsilon^{-2}$

At constant ρ these reduce to the corresponding original rates. The fixed-gap rate (61) is a separate cited special case; it is not obtained by extrapolating Theorem 27 beyond its hypotheses.

For the fixed uniform law ($\alpha = 1$), set $x = (a\sqrt{B})^{-1} = \rho^2$. The proved rate is of order $\rho^2 \log(1/\rho)$, whereas the proposed heuristic is of order $\rho^{5/2}\log(1/\rho)$. Thus failure of the heuristic also occurs for a single fixed μ.

The optimizer formula predicts an additional boundary at $\alpha = 2$: in a sufficiently large-budget regime its lower modulus has scale $(h/a)\sqrt{\rho \log(1/\rho)}$, versus $(h/a)\sqrt{\rho}$ for $\alpha > 2$. The above-two rate is matched in appendix B; at the quadratic law, appendix C proves the additional scale-entropy factor. Constants in the proved theorem need not be uniform as α approaches 1 or 2.

A.5 A certified general upper bracket, not a closed characterization

For any finite dyadic sequence $n_0, \dots, n_L = N$ with $4N \le B$, let

$$e_n = \mathbb{E}_\mu e_n(D), \quad v_0 = \rho + e_{n_0}, \quad v_j = e_{n_j} + e_{n_{j-1}}, \quad b_0 = 17\mathbb{E}_\mu m_{n_0}, \quad b_j = 34\mathbb{E}_\mu m_{n_j} \ (j \ge 1).$$

All these quantities are finite at finite caps. Put $S = \sum_{j=0}^{L} \sqrt{v_j b_j}$ and choose a finite categorical law $p_j \propto \sqrt{v_j/b_j}$. Sample Δ_J/p_J and apply (80) with cap N.

Proposition 31. *This estimator has expected work at most B and RMS error at most*

$$\sqrt{e_N^2 + S^2/N}.$$

Consequently the minimax risk is bounded above by the infimum of this expression over the finite grids, and also by $\rho/2$ and by the clipped original upper bound. The allocation here requires the stated integrals of μ or valid upper bounds for them.

Proof. The same coupling argument gives $\mathbb{E}\Delta_j^2 \le v_j$ and $\mathbb{E}W_j \le b_j$. Thus $\mathbb{E}Y^2 \le \sum v_j/p_j$ and $\mathbb{E}C \le \sum p_j b_j$. Their product equals at most S^2 for the stated choice of probabilities. Telescoping bounds the bias by e_N, and calibration bounds the variance by S^2/N. □

This bracket is constructive, but it does not remove the fixed-gap screening difficulty or establish arbitrary-law optimality. A sharp arbitrary-law replacement remains open; the matching-Ψ upper objective is false by theorem 20. It would be incorrect to present the variational identity as if it supplied that algorithm.

B Asymmetric screening above exponent two

The upper proof is uniform over the stated envelope and global rarity class. The matching lower proof cites the published fixed-gap input and supplies the simulation to variable magnitudes. This appendix remains a paper argument; the separate 713 kernel theorem is a different construction.

B.1 Claim and observational contract

For $\alpha > 2$ and $0 < a \le 1/2$, let $\mathcal{M}_{\alpha,a}$ be the probability laws on $(0, 1/2]$ satisfying

$$\mu((0, x]) \le (x/a)^\alpha \qquad (0 < x < a).$$

A coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$; its successive flips are conditionally independent Bernoulli variables with parameter $1/2 + (2S - 1)D$. Fresh coins are iid. All functions q below are measurable, $0 \le q \le 1$, and $\theta = \int q \, d\mu \le \rho$, where $0 < \rho \le 1/4$. Only actual coin flips are charged. Independent random bits and arithmetic are free, as in the repository's experiment.

An explicit realization uses mutually independent variables $D_i \sim \mu$, $U_i \sim \mathrm{Unif}(0, 1)$, and $V_{i,n} \sim \mathrm{Unif}(0, 1)$, for $i, n \ge 1$, on their countable product probability space. Set $S_i = \mathbf{1}_{\{U_i \le q(D_i)\}}$ and $X_{i,n} = \mathbf{1}_{\{V_{i,n} \le 1/2 + (2S_i - 1)D_i\}}$. An additional independent seed supplies every algorithmic random choice. These are measurable variables and give the required conditional iid flip tapes and independent fresh coins. The algorithms below use only the next unused $X_{i,n}$ on a requested coin. Their likelihoods, comparisons, and stopping decisions depend only on finite observed prefixes. The random-level and Poisson choices are measurable functions of the independent seed.

Set

$$\beta = 1 - 2/\alpha, \qquad A_0 = \frac{\alpha}{\alpha - 2}, \qquad A_1 = \frac{\alpha}{(\alpha - 2)^2},$$
$$w_\alpha = \frac{3}{2}\left[A_0\left(1 + 10\log 2 + \frac{1}{e\beta}\right) + A_1\right], \qquad K = \frac{153}{56} < 3.$$

All constants here precede the choice of a, μ, ρ, q, B.

Theorem 32 (Uniform upper theorem above two)**.** *For every $\alpha > 2$, $0 < a \le 1/2$, and $0 < \rho \le 1/4$, there is one observable single-coin estimator $Y = Y_{a,\rho}$, independent of μ and q, such that for every $\mu \in \mathcal{M}_{\alpha,a}$ and every admissible q,*

$$\mathbb{E}Y = \theta, \qquad \mathbb{E}Y^2 \le K\rho, \qquad \mathbb{E}T_Y \le w_\alpha/a^2.$$

In particular all integrability and finite-work assertions are conclusions. For every $B \ge 1$, a family of observable algorithms depending only on (α, a, ρ, B) has uniform expected work at most B and worst-case RMS at most

$$\min\left\{\frac{\rho}{2}, \sqrt{K w_\alpha}\,\frac{\sqrt{\rho}}{a\sqrt{B}}\right\}.$$

The final budgeted estimator need not be unbiased, because clipping or a constant estimate may be used. The underlying compound-Poisson estimator is unbiased.

The theorem gives a known-law result for the exact power law $F(d) = (d/a)^\alpha$ on $(0, a]$. Its proof also gives the stated uniform upper bound for the larger margin class. This is not an arbitrary-law solution and not an estimator independent of a rarity bound. In particular constants diverge as $\alpha \downarrow 2$.

B.2 Observable asymmetric sequential classifier

For $k = 0, 1, \ldots$ define known reference gaps and weights

$$r_k = a2^{-(k+1)}, \qquad \pi_k = 2^{-(k+1)}.$$

For the observed flips $X_1, X_2, \ldots$ of one coin, let

$$L^+_{k,n} = \prod_{i=1}^{n} (1 + 2r_k)^{X_i}(1 - 2r_k)^{1-X_i}, \quad L^-_{k,n} = \prod_{i=1}^{n} (1 - 2r_k)^{X_i}(1 + 2r_k)^{1-X_i}.$$

These are likelihood ratios against the fair-coin law. Fix $0 < \varepsilon \le 1$. At the first $n \ge 1$ for which some $k \ge 0$ satisfies either

$$L^-_{k,n} \ge \frac{1}{\pi_k \varepsilon} \quad \text{or} \quad L^+_{k,n} \ge \frac{1}{\pi_k \rho \varepsilon}, \tag{B.C}$$

stop and return $Z_\varepsilon = 0$ in the former case and $Z_\varepsilon = 1$ in the latter. Fix negative priority in case both conditions first occur together. The stopping time is T_ε.

This rule does not observe D, S, or q. It also does not require infinitely many operations at any finite observation prefix. If $k \ge n \ge 1$, then $2r_k \le 2^{-k}$ and

$$L^{\pm}_{k,n} \le (1 + 2r_k)^n \le \exp(n2^{-k}) \le e^{1/2} < 2, \qquad \frac{1}{\pi_k \varepsilon} \ge 2^{k+1} \ge 4.$$

Thus only the finite set $k < n$ needs to be checked at time n. The positive threshold is even larger. The products for any newly activated index can be computed from n and the observed head count. Every action and output is therefore a measurable function of the independent seed and observable history.

Lemma 33 (Errors and expected work)**.** *For every fixed $0 < d \le 1/2$ and $s \in \{0, 1\}$,*

$$\mathbb{P}(Z_\varepsilon = 0 \mid d, 1) \le \varepsilon, \qquad \mathbb{P}(Z_\varepsilon = 1 \mid d, 0) \le \rho\varepsilon.$$

Put $x = \min(d, a)$ and $c_0 = 1 + 2\log 2$. Then

$$\mathbb{E}(T_\varepsilon \mid d, s) \le \frac{3}{2} x^{-2} \left[c_0 + \log(a/x) + \log(1/\varepsilon) + s \log(1/\rho)\right]. \tag{B.1}$$

In particular this stopping time is finite almost surely for every allowed latent state, including $d = 1/2$.

Proof. Under a positive coin, the expected next factor of L^-_k is $1 - 4r_k d \le 1$; under a negative coin, that of L^+_k is the same. Consequently these wrong-sign processes are nonnegative supermartingales starting at one. For completeness, stopping a nonnegative supermartingale M at its first crossing of b and at a deterministic horizon N gives $b\mathbb{P}(\max_{n \le N} M_n \ge b) \le \mathbb{E}M_{\tau \wedge N} \le 1$. Letting $N \uparrow \infty$ proves the crossing bound $1/b$ without an unbounded optional-stopping assumption. Applying this to each k and summing $\sum_k \pi_k = 1$ proves the two error bounds, once almost-sure termination is established below.

Choose $k = \lceil \log_2(a/x) \rceil$. Then $x/4 < r_k \le x/2$ (the weaker $x/4 \le r_k \le x/2$ suffices), $r_k \le 1/4$, and $(k + 1)\log 2 \le \log(a/x) + 2\log 2$. Under the true sign, the logarithm of the corresponding correct-sign likelihood ratio is a sum of iid bounded increments, with mean

$$g(d, r_k) = \tfrac{1}{2}\log(1 - 4r_k^2) + d \log \frac{1 + 2r_k}{1 - 2r_k}.$$

The elementary inequalities $\log(1-z) \geq -z/(1-z)$ and $\log((1+u)/(1-u)) \geq 2u$ yield

$$g(d, r_k) \geq -\tfrac{8}{3} r_k^2 + 4dr_k \geq \tfrac{8}{3} x r_k \geq \tfrac{2}{3} x^2. \tag{B.2}$$

Every upward log increment is at most $\log(1+2r_k) < 1$. If σ is the first time this particular log process reaches

$$A = \log(1/\varepsilon) + (k+1)\log 2 + s\log(1/\rho),$$

then for every integer N, its value at $\sigma \wedge N$ is at most $A+1$. The elementary bounded-horizon identity

$$\mathbb{E} \sum_{i=1}^{\sigma \wedge N} \xi_i = \sum_{i=1}^{N} \mathbb{E}[\xi_i \mathbf{1}_{\{\sigma \geq i\}}] = g(d, r_k)\mathbb{E}(\sigma \wedge N)$$

uses only that $\{\sigma \geq i\}$ is determined before increment i. Hence $\mathbb{E}(\sigma \wedge N) \leq (A+1)/g(d, r_k)$. Monotone convergence proves $\mathbb{E}\sigma \leq (A+1)/g(d, r_k) < \infty$ and therefore $\sigma < \infty$ almost surely. Since $T_\varepsilon \leq \sigma$, equations (B.1)–(B.2) follow. This also completes the termination needed for the error assertions. □

B.3 The moment calculation that removes the logarithmic loss

The relevant distinction from symmetric high-confidence screening is that $\log(1/\rho)$ in (B.1) is paid only on positive coins.

Lemma 34 (Margin integration)**.** *If $\mu \in \mathcal{M}_{\alpha,a}$, $x = \min(D, a)$, $\alpha > 2$, and $0 \leq q \leq 1$, $\mathbb{E}q(D) \leq \rho$, then*

$$\begin{aligned} \mathbb{E}x^{-2} &\leq A_0 a^{-2}, \\ \mathbb{E}[x^{-2}\log(a/x)] &\leq A_1 a^{-2}, \\ \mathbb{E}[q(D)x^{-2}] &\leq A_0 a^{-2}\rho^\beta. \end{aligned} \tag{B.3}$$

For the exact power law the first two inequalities are equalities, and the last is attained by $q(d) = \mathbf{1}_{\{d \leq a\rho^{1/\alpha}\}}$.

Proof. The variable x stochastically dominates the power-law variable $aU^{1/\alpha}$, where U is uniform on $(0,1)$. Integrating decreasing nonnegative functions, or equivalently applying the layer-cake identity to their sublevel events, gives the first two inequalities. Their explicit integrals are

$$\frac{\alpha}{a^\alpha}\int_0^a d^{\alpha-3}\,dd = A_0 a^{-2}, \qquad \frac{\alpha}{a^\alpha}\int_0^a d^{\alpha-3}\log(a/d)\,dd = A_1 a^{-2}.$$

For the third, Tonelli's theorem and $q \leq 1$ give

$$\mathbb{E}[qx^{-2}] = a^{-2}\mathbb{E}q + \int_{a^{-2}}^{\infty} \mathbb{E}[q\mathbf{1}_{\{x^{-2}>u\}}]\,du \leq a^{-2}\rho + \int_{a^{-2}}^{\infty} \min\{\rho, (a^2u)^{-\alpha/2}\}\,du.$$

Splitting at $a^{-2}\rho^{-2/\alpha}$ evaluates the last expression as $A_0 a^{-2}\rho^{1-2/\alpha}$. The claimed equality case follows by direct integration. These arguments concern nonnegative extended integrals; finiteness is established by the displayed evaluations. □

Combining (B.1) and (B.3), with $\theta \leq \rho$, gives the uniform bound

$$\mathbb{E}T_\varepsilon \leq \frac{3}{2a^2}\left[A_0\{c_0 + \log(1/\varepsilon)\} + A_1 + A_0\rho^\beta\log(1/\rho)\right]. \tag{B.4}$$

Here $\rho^\beta\log(1/\rho) \leq 1/(e\beta)$, obtained by maximizing $te^{-\beta t}$ over $t \geq 0$. This is where $\alpha > 2$ is essential. The logarithm is controlled uniformly in the supplied rarity bound.

B.4 Unbiased random-level estimator

The single-term randomized telescoping device is an established unbiased multilevel ingredient; see Vihola [4, Theorem 3]. The particular classifier and the rarity-sensitive moment and work calculation below are analyzed explicitly. Let $\varepsilon_j = 16^{-(j+1)}$ for $j \geq 0$ and use the above classifiers $Z_j = Z_{\varepsilon_j}$ on one common coin-flip tape. Define

$$\Delta_0 = Z_0, \qquad \Delta_j = Z_j - Z_{j-1} \quad (j \geq 1).$$

Choose an independent J with $p_j = \mathbb{P}(J = j) = 2^{-(j+1)}$, draw one fresh coin, and return $Y = \Delta_J / p_J$. Only the classifiers actually needed for this increment are evaluated. Both threshold families in (B.C) grow by the same factor 16 between successive levels, so $T_j \geq T_{j-1}$ pathwise. The increment costs exactly T_J, with the earlier decision stored from the same prefix. No assumption about independence between classifier levels is made.

The conditional error bounds give

$$\mathbb{P}(Z_j \neq S) \leq (\theta + \rho)\varepsilon_j \leq 2\rho\varepsilon_j, \qquad \mathbb{E}Z_0^2 = \mathbb{E}Z_0 \leq \theta + \rho\varepsilon_0 \leq \tfrac{17}{16}\rho.$$

For $j \geq 1$, a disagreement implies an error of at least one classifier, hence

$$\mathbb{E}\Delta_j^2 \leq 2\rho(\varepsilon_j + \varepsilon_{j-1}) = \tfrac{17}{8}\rho\, 16^{-j}.$$

All square terms are nonnegative, so Tonelli gives

$$\mathbb{E}Y^2 = \sum_{j\geq 0} \frac{\mathbb{E}\Delta_j^2}{p_j} \leq \left(\frac{17}{8} + \frac{17}{4}\sum_{j\geq 1} 8^{-j}\right)\rho = \frac{153}{56}\rho. \tag{B.5}$$

This proves square integrability, hence integrability, before using signed expectations. Alternatively $\sum_j \mathbb{E}|\Delta_j| < \infty$ directly follows from the same disagreement bounds because $|\Delta_j| = \Delta_j^2$ for $j \geq 1$. Consequently

$$\mathbb{E}Y = \sum_{j\geq 0} \mathbb{E}\Delta_j = \lim_{j\to\infty} \mathbb{E}Z_j = \mathbb{E}S = \theta. \tag{B.6}$$

Finally (B.4), $\mathbb{E}(J+1) = 2$, and $\log(1/\varepsilon_j) = 4(j+1)\log 2$ imply

$$\mathbb{E}T_Y \leq \frac{3}{2a^2}\left[A_0\{1 + 10\log 2 + 1/(e\beta)\} + A_1\right] = w_\alpha / a^2. \tag{B.7}$$

Equations (B.5)–(B.7) prove the single-coin part of Theorem 32, with separate uniform second-moment and work bounds.

B.5 Exact expected-budget calibration and matching power-law rate

Independent Poisson replication and its variance formula are also established ingredients; see Vihola [4, Example 8]. We give the direct calculation to verify the exact expected budget, including when the expected count is below one. Set $\lambda = Ba^2/w_\alpha > 0$. Draw an independent $M \sim \mathrm{Poisson}(\lambda)$ and run independent copies $Y_1, \ldots, Y_M$ of the single-coin procedure on fresh coins. Return

$$\widehat{\theta} = \lambda^{-1}\sum_{i=1}^{M} Y_i.$$

The empty sum is zero. Conditioning first on M, then using its mean and variance λ, proves

$$\mathbb{E}\widehat{\theta} = \theta, \quad \mathrm{Var}(\widehat{\theta}) = \lambda^{-1}\mathbb{E}Y^2 \leq \frac{K w_\alpha \rho}{a^2 B}, \quad \mathbb{E}T \leq \lambda w_\alpha / a^2 = B.$$

All sums stop almost surely and have finite expected work. These identities are valid even when $\lambda < 1$; no rounding of the budget is hidden. Clip the result to $[0, \rho]$, which cannot increase squared error for $\theta \in [0, \rho]$. When its displayed RMS bound exceeds $\rho/2$, instead output $\rho/2$ at zero work. The choice uses only the allowed parameters. This proves the budgeted part of Theorem 32.

Lemma 35 (Localized fixed-gap RMS lower bound)**.** *There is a universal $c > 0$ such that the minimax RMS risk for fixed gap $0 < a \le 1/2$, prior $0 \le \vartheta \le \rho \le 1/4$, and uniform expected work at most $B \ge 1$ is at least $c \min\{\rho, \sqrt{\rho}/(a\sqrt{B})\}$. This lemma uses Lee–Valiant Corollary 19 and Lemma 18 as external inputs.*

Proof. The exact imported assertions imply the following: there are universal $a_*, c_*, C_* > 0$ such that, if $a < a_*$ and $u, v, v/u < c_*$, the KL divergence of any finite-prefix fully adaptive fixed-gap transcript under sign priors u and $u + v$ is at most $C_* v^2 a^2 \mathbb{E}_u T/u$. Indeed, Corollary 19 bounds each single-coin hybrid's squared Hellinger distance by a universal multiple of v^2a^2/u times its expected observations under u. Lemma 18 bounds joint KL by the sum of those distances. Summing coin work counts each observation once. Their Hellinger normalization is absorbed in C_*. We use the original-prior, not midpoint-prior, expectation. All their smallness hypotheses are enforced in the choices below.

For arbitrary almost-surely terminating policies with finite expected work, first stop at $T \wedge n$ and number coins by first observation. At most n coins are then observed, so the finite-prefix statement applies. This numbering preserves the paper's creation and interleaving model: an unobserved coin has provided no information, and its iid latent draw can be deferred until its first observation. A dictionary from the policy's original coin labels to the new indices preserves every revisit. Unobserved-label choices use only the seed and observed history, so this relabeling changes neither their law nor any charged observation or output. Include the independent random seed in the transcript; conditioning on it and averaging extends the deterministic statement. Relative entropy along increasing prefix sigma fields converges to full transcript relative entropy. The stopped output is measurable in this sigma field. Thus the same bound holds at unbounded stopping, without a globally fixed finite coin family.

Write $r = \min\{\rho, \sqrt{\rho}/(a\sqrt{B})\}$, and take $u = b\rho$, $v = \gamma r$ for fixed sufficiently small positive universal constants b, γ. Choose $b + \gamma \le 1$ and $\gamma/b < c_*$, so $u, u + v \in [0, \rho]$ and all prior smallness assumptions hold. Since $r^2a^2B \le \rho$, for $a < a_*$ the transcript KL is at most $C_*\gamma^2/b$.

For $a \ge a_*$, give the algorithm each coin's true sign on its first observation. This more informative experiment simulates all requested flips from that sign. Each newly observed sign has Bernoulli KL at most $2v^2/u$, using $u + v \le 1/4$ and the KL–chi-square inequality. The expected number of newly observed signs is at most $\mathbb{E}_u T \le B$, so full KL is at most $2v^2B/u \le 2\gamma^2/(ba_*^2)$. The same finite-prefix argument justifies unbounded stopping. This also covers the endpoint $a = 1/2$.

Choose γ small enough that both displayed KL bounds are at most $1/8$. Pinsker's inequality then bounds total variation by $1/4$. If an estimator has RMS at most R at both priors, thresholding halfway between them gives a test whose two error probabilities sum to at most $8R^2/v^2$. Every test has error sum at least $1 - \mathrm{TV} \ge 3/4$. Therefore $R \ge v/4$, proving the claim with $c = \gamma/4$. This proves all budget and rarity regimes, including the saturated regime, from the precise source inputs rather than an informal high-confidence asymptotic statement. □

Corollary 36 (Exact-power regime $\alpha > 2$)**.** *Let μ have distribution function $(d/a)^\alpha$ on $(0, a]$ with $\alpha > 2$. Subject to the published fixed-gap lower bound of Lee–Valiant, there is a universal $c > 0$ and, for each $\alpha > 2$, a finite C_α such that for all $0 < a \le 1/2$, $0 < \rho \le 1/4$, and $B \ge 1$,*

$$c \min\left\{\rho, \frac{\sqrt{\rho}}{a\sqrt{B}}\right\} \le R_{\mu,\rho}(B) \le C_\alpha \min\left\{\rho, \frac{\sqrt{\rho}}{a\sqrt{B}}\right\}.$$

Moreover this risk is comparable, with constants depending only on α, to $\Psi_{\mu,\rho}(B^{-1/2})$.

Proof. The upper bound was just proved. For the lower bound it is enough to restrict to constant sign functions $q(d) = \vartheta \in [0, \rho]$. An experiment using fixed-gap a coins can simulate the variable-gap experiment at the same charged work: sample $D \sim \mu$ privately for each fresh fixed-gap coin; on each request, flip that coin once, and with probability D/a return its flip, otherwise return an independent fair bit. Conditional on the fixed coin's sign and its privately sampled D, the returned flips are iid Bernoulli$(1/2 \pm D)$. The same latent D is retained on revisits, and is not disclosed to the simulated algorithm. The simulation uses one charged flip per returned observation even when that flip is discarded. It preserves all interleaving and stopping choices. Thus a smaller variable-gap risk would contradict Lemma 35. This reduction does not need the continuation note's arbitrary-law lower-bound theorem.

For comparison with the functional, constant allocation $w \equiv \rho$ and $D \le a$ give

$$\Psi_{\mu,\rho}(h) \ge \min\{\rho, h\sqrt{\rho}/a\}.$$

For every feasible w, Cauchy–Schwarz and $\mathbb{E}D^{-2} = A_0/a^2$ give

$$\int \min\{w, h\sqrt{w}/D\}\, d\mu \le \min\{\rho, h\sqrt{\rho A_0}/a\}.$$

Taking the supremum proves the claimed comparison, including saturation. □

B.6 Scope, edge cases, and failed approaches

The classifier and estimator operate at $D = 1/2$, with no likelihood logarithm at zero because the reference gaps satisfy $r_k \le 1/4$. At arbitrarily small $D > 0$ each conditional expected stopping time is finite; the margin assumption then proves population integrability. The level J has unbounded support and finite mean. For the admissible law $\mu = \delta_{1/2}$, $a = 1/2$, and $q = 0$, all observations are zero. At time n every correct-sign likelihood is at most $(3/2)^n$, whereas every level-j correct-sign threshold is at least $2/\varepsilon_j$. Thus each fixed n is exceeded for sufficiently large j. Since all $p_j > 0$, actual work has genuinely unbounded support and finite mean, not merely an unused unbounded seed. A finite Poisson number of such procedures also has finite expected work.

The global constraint $\int q\, d\mu \le \rho$ is used precisely in (B.3); no pointwise bound $q \le \rho$ is imposed. Its extremizer puts all positive mass on the smallest gaps. The algorithm never assumes known positive or negative conditional magnitude laws. The result is uniform jointly in B, ρ, a for each fixed $\alpha > 2$, but is not uniform as $\alpha \downarrow 2$.

Two earlier approaches exposed the obstruction addressed here. Symmetric high-confidence screening charges $\log(1/\rho)$ on all coins, retaining an unwanted logarithm. An unamplified ballot-survival estimator has positive conditional second moment of order $1/D$; placing positives near the threshold then gives an unwanted $\rho^{1-1/\alpha}$ moment scale. The asymmetric classifier followed by an unbiased random-level correction avoids both problems. The negative classifier error is $\rho\varepsilon$, whereas the positive error is only ε; the extra positive work is absorbed by the sharp global-rarity integral in (B.3).

The same proof does not settle $\alpha = 2$: its inverse-square integral diverges, and the uniform control of $\rho^\beta \log(1/\rho)$ loses meaning at $\beta = 0$. It does not characterize arbitrary laws with a finite inverse-square moment, because that moment alone does not uniformly control the logarithmically weighted moment or the rarity-weighted cost. It does not refute that larger conjecture. A direct finite-support modification gives finite bounds with support-dependent constants, but this is not the universal two-atom theorem.

C The full critical quadratic law

This appendix gives the full paper argument, including the upper bound. Its finite-gap sign-entropy ingredient is Lean-checked in the model stated in section 11. The continuum null-cost theorem, statistical common pair, and population transfer remain unformalized. Existing agent review is not independent specialist verification.

C.1 Experiment, quantifiers, and result

Throughout fix $a = 1/4$ and

$$\mu(dd) = \frac{2d}{a^2}\mathbf{1}_{(0,a]}(d)\,dd.$$

For $0 < \rho \le 1/4$, let

$$\mathcal{Q}_\rho = \{q : (0,a] \to [0,1] \text{ measurable} : \int q\,d\mu \le \rho\}, \qquad \theta(q) = \int q\,d\mu.$$

Each genuine fresh coin has an independent hidden magnitude $D \sim \mu$, hidden sign $S \mid D \sim \mathrm{Ber}(q(C.D))$, and conditionally iid flips of mean $1/2 + (2S-1)D$. Neither latent variable is observed. One realization uses independent uniforms on $(0,1)$ for each coin: $D = a\sqrt{U_0}$, $S = \mathbf{1}_{\{U_1 \le q(C.D)\}}$, and flips $X_k = \mathbf{1}_{\{U_{k+1} \le 1/2+(2S-1)D\}}$ for $k \ge 1$. Measurable functions equal μ-almost everywhere define the same experiment; Borel versions may be used in this standard probability space.

The algorithm has an independent seed and sees only its seed and the observed history. At each step it stops and returns a real measurable output, revisits an observed coin, or observes a fresh coin. Every flip costs one. The output may be biased and nonlinear, and the choices may depend on all previous observations. Stopping may be unbounded. A design may know μ, ρ, B but is fixed before the unknown q is supplied. Work $T = \infty$ on nontermination. Define

$$R_{\mu,\rho}(C.B) = \inf_{A:\,\sup_{q\in\mathcal{Q}_\rho}\mathbb{E}_q T_A \le B} \sup_{q\in\mathcal{Q}_\rho} \left(\mathbb{E}_q(\widehat{\theta}_A - \theta(q))^2\right)^{1/2}.$$

This class permits arbitrary creation and interleaving, not a globally fixed number of coins. We number coins by their first charged observation. This is equivalent to giving names to iid unseen coins earlier: an unused coin's latent state is independent of the observed history and seed, and choosing among unused names using only that information still selects an iid fresh state. The relabeling retains the map for every later revisit. Creating or discarding an unobserved coin supplies no information. Consequently at most K different coins occur before the Kth charged observation.

Write $L = 1 + \log(1/\rho)$, $H = 1 + \log L$, and $J = \rho L H$. For the lower proof we use the convex subclass

$$\mathcal{X}_\rho = \{q \in L^\infty(\mu) : 0 \le q \le 1/2 \ \mu\text{-a.e.}, \ \int q\,d\mu \le \rho\}.$$

This pointwise cap restricts the lower-bound adversary only; the algorithmic upper and minimax definition still use the full global-rarity class.

Theorem 37 (Full critical law)**.** *There exist universal constants $0 < c < C < \infty$ such that, for the one fixed law above, every $0 < \rho \le 1/4$ and $B \ge 1$ satisfy*

$$c\min\{\rho, \sqrt{J/B}\} \le R_{\mu,\rho}(C.B) \le C\min\{\rho, \sqrt{J/B}\}. \tag{C.A}$$

The lower bound holds for every observable algorithm in the stated full experiment, including arbitrary randomization, nonlinear estimation and data-dependent creation, revisiting and interleaving. The upper permits the known rarity and law; it has separately finite uniform work and squared error.

The upper construction is reproduced in the final subsection. Its source and the inherited proof identities are recorded in the v5 source manifest. The new lower proof has three steps. First, sign-testing entropy forces a small-bias variance product using *all-negative work*, not a supremum of work over populations. Second, convex estimation and minimax supply one hard pair valid for every capped single-coin observation policy. Third, a midpoint simulation charges those bounds to the actual work of an interactive population algorithm. The last step is the established Lee–Valiant Hellinger decomposition mechanism [1, Lemma 17], proved below at finite prefixes to make its observation and stopping scope explicit.

The convex-estimation step follows Juditsky–Nemirovski's saddle-point framework [17]; the passage from its exponential tails to RMS risk is written out here. Polyanskiy–Wu [18] already establish broad quadratic-risk affine comparisons and convex duality for functional estimation. Neither that general principle nor the Hellinger decomposition is claimed as new. The cost-sensitive common pair and its application to this critical hidden-coin class are the proposed contribution; priority remains unverified.

C.2 A single-coin entropy bound with all-negative reference cost

Fix $a = 1/4$ and $\mu(dd) = 2da^{-2}\mathbf{1}_{(0,a]}(d)\,dd$. Throughout this section,

$$\mathcal{X}_\rho = \left\{q : 0 \le q \le \tfrac{1}{2},\ \int q\,d\mu \le \rho\right\}, \qquad 0 < \rho \le \tfrac{1}{4}, \qquad L = 1 + \log(1/\rho), \quad H = 1 + \log L.$$

All functions defining populations are measurable. A single genuine coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$; conditionally on (D, S) its observations are iid Bernoulli with parameter $1/2 + (2S-1)D$. A policy receives only its independent seed and observed bits. It may know μ, a, ρ, but not q, D, S.

Theorem 38 (The critical product with null-population cost)**.** *There is a universal $c_0 > 0$ such that the following holds for every $0 < \rho \le 1/4$. Let an observable single-coin procedure have real output Y and stopping time $T \in \mathbb{N}_0 \cup \{\infty\}$, and suppose*

$$\sup_{q\in\mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64, \qquad V := \sup_{q\in\mathcal{X}_\rho} \mathrm{Var}_q\, Y < \infty, \qquad \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q T < \infty, \qquad \theta(q) = \int q\,d\mu.$$

Writing $W_0 = \mathbb{E}_{q=0}T$, one has

$$VW_0 \ge c_0 \rho L H. \tag{C.BC1}$$

The same order lower bound holds with $M = \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Y^2$ in place of V. The cost is the expectation under the all-negative population, rather than the supremum of expected work over the parameter class. The policy may stop before observing the coin and may have unbounded stopping.

The proof adapts the separately documented critical-two single-coin argument. We reproduce its finite-gap dependency to make this statement self-contained. Its entropy mechanism follows Chen–Li–Qiao, arXiv:1608.06031, Lemma 4.1; the Bernoulli, one-sided-cost statement used here is proved explicitly below. No novelty is claimed for that mechanism.

Lemma 39 (Finite-gap sign entropy)**.** *There are universal $c_s > 0$ and $K_s < \infty$ with the following property. Let $d_1 > \cdots > d_K > 0$ satisfy $d_j/d_{j+1} \ge 2$ and $d_j \le 1/128$. Suppose one observable sign-testing procedure, the same at all these gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$, and finite expected stopping time τ under both signs. For $K \ge K_s$,*

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j}\tau \ge c_s K \log K. \tag{C.BC2}$$

Correctness and termination under the fair coin are not required. One may take $c_s = 1/512$ *and* $K_s = \lceil e^{32} \rceil$.

Proof. Put $b_0 = 1/1024$, $n_j = \lfloor b_0/d_j^2 \rfloor$, $u_j = \lceil 4\mathbb{E}_{-d_j}\tau \rceil$, $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$, and $\bar{c} = K^{-1}\sum_j c_j$. Since $d_j \le 1/128$, $n_j \ge b_0/(2d_j^2) \ge 1$. The event of returning minus by time n_j is measurable in the independent seed and the first n_j flips, including unused coordinates if the test has stopped. The KL divergence between these full prefixes at the minus and plus laws is at most

$$n_j \frac{(2d_j)^2}{(1/2+d_j)(1/2-d_j)} \le \frac{64}{3} n_j d_j^2 \le \frac{1}{48}.$$

Bernoulli KL is at most its chi-square divergence. Pinsker's inequality bounds the total variation by less than $1/8$. Since an early minus is an error under the plus law, it has probability at most $1/10 + 1/8$ under the minus law. Markov's inequality gives $\mathbb{P}_{-d_j}\{\tau > u_j\} \le 1/4$. Consequently

$$A_j = \{\text{output minus and } n_j < \tau \le u_j\} \quad \text{satisfies} \quad \mathbb{P}_{-d_j}(A_j) \ge 1 - \tfrac{1}{10} - (\tfrac{1}{10} + \tfrac{1}{8}) - \tfrac{1}{4} > \tfrac{1}{4}.$$

Let P_0 denote the fair-coin law with the same seed and policy. Comparing the deterministic prefixes of length u_j, their KL divergence from minus to fair is at most $4d_j^2 u_j \le 16c_j + 1$. For an event of probability $p \ge 1/4$ under the minus law and probability z under the fair law, binary data processing and $\mathrm{kl}(p,z) \ge p\log(1/z) - \log 2$ therefore give

$$P_0(A_j) \ge e^{-64c_j - 8}. \tag{C.BC3}$$

Only deterministic finite prefixes have been compared; this does not invoke optional stopping under a possibly nonterminating fair law.

At least $K/2$ indices form a set G with $c_j \le 2\bar{c}$. For them,

$$(n_j, u_j] \subset [b_0/(2d_j^2), (8\bar{c}+1)/d_j^2].$$

The scales d_j^{-2} have successive ratios at least four. At any finite time the number of these intervals containing that time is at most $1 + \log_4((16\bar{c}+2)/b_0)$. All events A_j refer to the same terminal time and output on a realization under the fair law; on a nonterminating path none occurs. Summing their indicators and using (C.BC3) yields

$$(K/2)e^{-128\bar{c}-8} \le 1 + \log_4(1024(16\bar{c}+2)). \tag{C.BC4}$$

If $x = \log K \ge 32$ and $\bar{c} < x/512$, the left side exceeds $\frac{1}{2}e^{3x/4-8} \ge \frac{1}{2}e^{x/2} \ge x^2/16$. The right side is less than $1 + \log_4(32x + 2048) < 6 + x/4$, contradicting $x \ge 32$. Thus $\bar{c} \ge (\log K)/512$, as claimed. □

Proof of Theorem 38. The bias bound implies $M \le V + (65\rho/64)^2 < \infty$. First define the nonnegative extended conditional moments

$$M_s(d) = \mathbb{E}_{d,s} Y^2, \qquad w_s(d) = \mathbb{E}_{d,s} T.$$

The admissible populations $q = 0$ and $q \equiv \rho$ imply $\int M_- d\mu \le M$, $\int M_+ d\mu \le M/\rho$, and finite integrals of both work functions. All conditional quantities used below are thus finite outside a common null set. Define signed conditional means there, and give them arbitrary values on that null set. For $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$, the two bias bounds imply

$$\int f \, d\mu = (\mathbb{E}_\rho Y - \mathbb{E}_0 Y)/\rho \ge 31/32.$$

Conditioning on the latent magnitude and then on its sign at the constant-prior population gives

$$V \geq \mathrm{Var}_\rho(Y) \geq \rho(1-\rho)\int f^2\,d\mu \geq \rho(1-\rho)(31/32)^2 \geq 2\rho/3. \tag{C.BC5}$$

This uses only the bias guarantee, not a pointwise unbiased response. It also shows

$$M \leq \tfrac{3}{2}V. \tag{C.BC6}$$

Normalize the first observation. The event $A = \{T \geq 1\}$ is decided by the independent seed before any observation, so its probability p is independent of the latent state. It is positive: otherwise the expectations at target values zero and ρ could not both satisfy the bias bound. The finite constant $c = \mathbb{E}[Y\mathbf{1}_{A^c}]$ is also population independent. Under the seed law conditioned on A, form the active procedure

$$Z = pY + c, \qquad T_Z = T.$$

It preserves every expectation of Y and always takes at least one observation. The conditioning is implementable by resampling independent seeds before observing the coin; its constants depend on the specified policy and known laws, not on q, D, S.

For $p < 1$, expansion, Cauchy–Schwarz on A^c, and completing the square give

$$\begin{aligned}\mathbb{E}_q Z^2 &= p\mathbb{E}_q[Y^2\mathbf{1}_A] + 2c\mathbb{E}_q Y - c^2\\ &\leq pM + 2c\mathbb{E}_q Y - c^2/(1-p)\\ &\leq pM + (1-p)(\mathbb{E}_q Y)^2 \leq m := pM + 2\rho^2.\end{aligned}$$

For $p = 1$, take $c = 0$ and the final bound holds directly. Apply (C.BC5) to Z to obtain $m \geq 2\rho/3$ and $pM \geq 2\rho/3 - 2\rho^2 \geq \rho/6$. Set $P = MW_0$ and $W_{0Z} = \mathbb{E}_0 T_Z = W_0/p \geq 1$. Then

$$m \leq 4pM \leq 4P, \qquad mW_{0Z} \leq 4P, \qquad P \geq \rho/6. \tag{C.BC7}$$

Indeed $2\rho^2 \leq \rho/2 \leq 3pM$. Every work quantity in (C.BC7) is all-negative work.

Select good gaps within the capped global class. Now use conditional means and moments for Z, keeping the same notation. The bad set $B_- := \{M_- > 1/64\}$ has mass at most $64m$. The set

$$B_+ := \{M_+ - M_- > 2m/\rho\}$$

has mass less than 2ρ. Otherwise nonatomicity supplies $E \subset B_+$ of mass 2ρ, and the admissible $q = \frac{1}{2}\mathbf{1}_E$ gives

$$\mathbb{E}_q Z^2 = \int M_- d\mu + \tfrac{1}{2}\int_E (M_+ - M_-)d\mu > 2m,$$

contradicting its bound. The response bad set $B_f := \{f < 3/4\}$ also has mass less than 2ρ. If not, choose such an E of mass 2ρ. Its population has target ρ, but

$$\mathbb{E}_{\mathbf{1}_E/2} Z - \mathbb{E}_0 Z = \tfrac{1}{2}\int_E f\,d\mu < 3\rho/4,$$

whereas the bias bounds force at least $31\rho/32$.

Remove these bad sets and the conditional integrability null sets. The remaining good set G has $\mu(G^c) \leq 64m + 4\rho \leq 70m$. On it,

$$|\mathbb{E}_- Z| \leq 1/8, \qquad \mathbb{E}_+ Z \geq 5/8, \qquad M_- \leq 1/64, \qquad M_+ \leq 1/64 + 2m/\rho. \tag{C.BC8}$$

The use of sets of mass 2ρ is necessary for the cap $q \leq 1/2$; the argument does not substitute a pointwise bound $q \leq \rho$.

Build a common sign test. Let $n = \lceil 512(1 + m/\rho) \rceil \le 1282m/\rho$, the inequality using $m/\rho \ge 2/3$. On the same fixed-gap coin, restart the active procedure n times with independent seeds and consecutive unused flip blocks. Output plus if their mean exceeds 3/8. At a fixed latent state the restarts are iid: condition on each finite stopping prefix, use the independence of its unused iid suffix, and sum over the countably many prefixes. Finite conditional expected work ensures almost-sure termination for both signs at each good gap. No new latent magnitude is drawn. The same test is used at every good gap. By (C.BC8), either mean is at least 1/4 from the threshold, and Chebyshev bounds either error by

$$\frac{16(1/64 + 2m/\rho)}{n} < 1/10.$$

Its negative expected work at gap d is exactly $nw_-(d)$.

Average negative work over separated annuli. For $j \ge 3$, put

$$I_j = (a4^{-j}/2, a4^{-j}], \qquad \mu(I_j) = \tfrac{3}{4}16^{-j}, \qquad J = \left\lfloor \log_{16} \frac{3}{560m} \right\rfloor, \qquad K = J - 2.$$

When $J \ge 3$, every retained annulus has mass at least $140m$, so G occupies at least half of it. Choose $d_j \in G \cap I_j$ with

$$w_-(d_j) \le \frac{4}{\mu(I_j)} \int_{I_j} w_- \, d\mu.$$

This follows by averaging; the integral is positive since the active procedure always observes at least once. The choices are existential proof objects, not inputs to the policy. Successive gap ratios exceed two, every gap is at most 1/256, and

$$\sum_{j=3}^{J} d_j^2 w_-(d_j) \le \frac{16a^2}{3} \int w_- d\mu = \frac{16a^2}{3} W_{0Z}.$$

Lemma 39, the restart bound, and (C.BC7) imply, for $K \ge K_s$,

$$c_s K \log K \le \sum_{j=3}^{J} d_j^2 n w_-(d_j) \le \frac{16a^2}{3} \frac{1282m}{\rho} W_{0Z} \le \frac{30000a^2 P}{\rho}. \tag{C.BC9}$$

The last numerical constant is conservative, since $(16/3) \cdot 1282 \cdot 4 < 30000$.

Obtain one positive constant for every rarity. For sufficiently small ρ, if $P \ge \sqrt{\rho}$ the desired order already follows because $\sqrt{\rho} LH \to 0$. Otherwise (C.BC7) gives $m \le 4\sqrt{\rho}$ and

$$K \ge \frac{\frac{1}{2}\log(1/\rho) - \log(2240/3)}{\log 16} - 3.$$

For all sufficiently small universal ρ this exceeds K_s, is at least a positive universal multiple of L, and satisfies $\log K \ge \frac{1}{2}\log L$. Equation (C.BC9), with the fixed $a = 1/4$, gives $P \ge c\rho LH$. On the remaining compact interval of rarities, the lower bound $P \ge \rho/6$ in (C.BC7) allows the same conclusion after decreasing one positive universal constant. Finally (C.BC6) transfers the raw-moment lower bound to VW_0, proving (C.BC1). □

The all-negative cost is the strengthened part of this statement. It does not interchange separate suprema, assert a same-population product, or by itself reduce an arbitrary population algorithm to a single coin.

C.3 An RMS comparison for finite convex experiments

Let X be a finite-dimensional nonempty compact convex set and $\ell(x)$ a linear functional. Consider either of the following experiments:

1. independent finite categorical observations, whose positive probability vectors $p_i(x)$ are affine functions of x;
2. a finite vector of independent Poisson counts, whose strictly positive intensities $v_j(x)$ are affine functions of x.

In the first case let $\mathcal{F}$ be the finite-dimensional space of functions $c+\sum_i \phi_i(z_i)$. In the second let $\mathcal{F}=\{c+\sum_j \beta_j z_j\}$. Let P_x denote the whole observation law. All exponential moments of functions in $\mathcal{F}$ are finite. For fixed $\phi\in\mathcal{F}$, the function $x\mapsto \log \mathbb{E}_x e^{\phi}$ is concave: it is a sum of logarithms of positive affine forms in the categorical case, and an affine function of the intensities in the Poisson case. Moreover $\log(dP_x/dP_y)\in\mathcal{F}$.

Lemma 40 (RMS affine comparison)**.** *For any measurable estimator G with $r_G = \sup_{x\in X}(\mathbb{E}_x(G-\ell(x))^2)^{1/2}<\infty$ and any $\eta>0$, there exists $F\in\mathcal{F}$ such that*

$$\sup_{x\in X}(\mathbb{E}_x(F-\ell(x))^2)^{1/2}\le 12 r_G+\eta. \tag{C.2}$$

The result permits nonattainment of the infimum and imposes no restriction on the original estimator's output range.

Proof. Put $s=\log 2$. For $\alpha>0$ and $\phi\in\mathcal{F}$ define

$$\Phi(x,y;\phi,\alpha)=\ell(x)-\ell(y)+\alpha\log\mathbb{E}_y e^{\phi/\alpha}+\alpha\log\mathbb{E}_x e^{-\phi/\alpha}+2\alpha s,$$
$$2S=\inf_{\phi,\alpha>0}\max_{x,y\in X}\Phi(x,y;\phi,\alpha).$$

This real-valued continuous function is convex in (ϕ,α) by the perspective transformation and concave in (x,y) by the preceding log-moment observation. Sion's minimax theorem applies because $X\times X$ is compact convex and the other domain is convex. Thus the infimum and supremum commute. For fixed x,y, Cauchy–Schwarz gives

$$\inf_{\psi\in\mathcal{F}}\left(\log\mathbb{E}_y e^{\psi}+\log\mathbb{E}_x e^{-\psi}\right)=2\log\mathrm{Aff}(P_x,P_y),$$

with equality at $\psi=\frac12\log(dP_x/dP_y)$. Taking the infimum over $\alpha>0$ after putting $\phi=\alpha\psi$ therefore proves

$$2S=\max\{\ell(x)-\ell(y): x,y\in X,\ \mathrm{Aff}(P_x,P_y)\ge 1/2\}. \tag{C.3}$$

The constraint is closed and nonempty, so the maximum exists. Continuity of affinity is immediate from the finite categorical formula and, for Poisson vectors, from $\mathrm{Aff}(P_x,P_y)=\exp[-\frac12\sum_j(\sqrt{v_j(x)}-\sqrt{v_j(y)})^2]$. In particular $0\le S<\infty$.

For the maximizing pair in (C.3), total variation is at most $\sqrt{3}/2$ by the standard Cauchy–Schwarz affinity bound $\|P-Q\|_{\mathrm{TV}}\le\sqrt{1-\mathrm{Aff}(P,Q)^2}$. Testing the two endpoints by thresholding G at their functional midpoint gives, with $\Delta=2S$,

$$2r_G^2\ge\frac{\Delta^2}{4}(1-\|P_x-P_y\|_{\mathrm{TV}}),\qquad S\le\sqrt{\frac{2}{1-\sqrt{3}/2}}\,r_G. \tag{C.4}$$

Indeed each wrong decision has squared estimation error at least $\Delta^2/4$, and the sum of the two testing errors is at least one minus total variation. The case $S = 0$ requires no division by S.

Fix $R > S$. Choose (ϕ, α) with $\max_{x,y} \Phi \le 2R$. On the diagonal, Cauchy–Schwarz gives $\Phi(x, x; \phi, \alpha) \ge 2\alpha s$, hence $\alpha \le R/s$. Set

$$U = \max_x \{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x \{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \le 2R$. The additive estimator $F = \phi + (U - V)/2$ satisfies, uniformly in x,

$$\mathbb{E}_x e^{(F-\ell(x))/\alpha} \le e^{R/\alpha - s}, \qquad \mathbb{E}_x e^{(\ell(x)-F)/\alpha} \le e^{R/\alpha - s}.$$

For $t \ge 0$, Markov's inequality consequently gives $\mathbb{P}_x(|F - \ell(x)| > R + t) \le e^{-t/\alpha}$, since $2e^{-s} = 1$. Integrating this tail, rather than substituting a confidence guarantee for an RMS bound, yields

$$\mathbb{E}_x(F - \ell(x))^2 \le R^2 + 2R\alpha + 2\alpha^2 \le R^2 \left(1 + \frac{2}{s} + \frac{2}{s^2}\right). \tag{C.5}$$

The product of the squared constants in (C.4)–(C.5) is less than 144: $\log 2 > 2/3$ and $\sqrt{3}/2 < 7/8$ bound it by $8.5 \cdot 16 = 136$. Letting $R \downarrow S$, with the slack chosen for the given η, proves (C.2). This also proves uniform square integrability of the produced estimator, rather than assuming it. □

C.4 From a base-cost obstruction to one pair hard for every capped policy

In this section only, restrict the parameter set to

$$\mathcal{X}_\rho = \{q : 0 \le q \le 1/2, \int q \, d\mu \le \rho\}, \qquad J = \rho L H.$$

It is a weak-star compact convex subset of $L^\infty(\mu)$. The following implication separates its one required entropy input from the convex-experiment argument. The input is the base-cost bound proved in the accompanying base-cost section of the full-model note:

There is a universal $c_b > 0$ such that every observable single-coin estimator Y with $\sup_{q \in \mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64$, $V = \sup_{q \in \mathcal{X}_\rho} \mathrm{Var}_q Y < \infty$, $\sup_{q \in \mathcal{X}_\rho} \mathbb{E}_q T < \infty$ satisfies $V \mathbb{E}_0 T \ge c_b J$. (C.BC)

The cost in (C.BC) is at the all-negative population $q = 0$, not the supremum over populations. The two lemmas below are conditional on precisely this input. No algorithm-level lower bound is inferred from a variational optimization alone. Write $h(P, Q) = 1 - \mathrm{Aff}(P, Q)$ for squared Hellinger distance in the normalization used here; it takes values in $[0, 1]$.

Exact finite observation reduction

A capped local policy A includes its independent seed and stops after at most $m < \infty$ flips of one genuine coin. Its full observed transcript contains the seed and the stopped word. Its seed distribution and policy may depend on known parameters, but not on the actual q. For each word $x \in \{0, 1\}^t$, let $B_{t,x}$ be the measurable seed set that causes stopping exactly at that word. For $0 \le k \le t \le m$ put

$$a_{t,k} = \sum_{x:\, |x|=t,\, \#1(x)=k} \mathbb{P}(\text{seed} \in B_{t,x}),$$

and, with $p_s(d) = 1/2 + (2s-1)d$, put

$$m_{t,k}(q) = \int \left[(1-q(d))p_0(d)^k(1-p_0(d))^{t-k} + q(d)p_1(d)^k(1-p_1(d))^{t-k}\right] d\mu.$$

The finite statistic (T, K), where K is the number of observed heads, has probabilities $a_{t,k}m_{t,k}(q)$. Conditional on (T, K), the stopped word and seed have a distribution independent of q: their common likelihood multiplier is exactly $m_{t,k}(q)$ and cancels. This statement also follows directly by integrating any bounded transcript function over the finitely many word/seed cells, so it does not assume a latent oracle. Consequently the full transcript and this finite statistic have exactly the same Hellinger affinities for all pairs of populations.

Remove coordinates with $a_{t,k} = 0$. The remaining coordinates are positive for every q, since $p_s(d) \in [1/4, 3/4]$. They are affine weak-star continuous functions of q. Thus their probability vector together with $\theta(q)$ has a compact convex finite-dimensional range. In particular, for each capped A, its full-transcript squared Hellinger distance $h_A(q_-, q_+)$ is a continuous convex function of the pair on $\mathcal{X}_\rho^2$. Convexity follows from joint concavity of affinity for two affine probability vectors. This reduction retains the stopping time and expected cost, not just a fixed number of sampled bits.

Lemma 41 (A hard pair for each capped policy)**.** *Assume (C.BC). There are universal constants $c_\delta > 0$ and $C_H < \infty$ such that, for every $\rho \in (0, 1/4]$, every $0 < \delta \le c_\delta \rho$, and every randomized capped local policy A, there exist $q_-, q_+ \in \mathcal{X}_\rho$ with $\theta(q_+) - \theta(q_-) \ge \delta$ and*

$$h_A(q_-, q_+) \le C_H \frac{\delta^2}{J} \mathbb{E}_0 T_A. \tag{C.6}$$

The full observed seed and stopped history are included in the Hellinger distance.

Proof. Let $w = \mathbb{E}_0 T_A$. If $w = 0$, the seed-only event of taking a first observation has probability zero, so A has the same transcript law for every q. The pair $q_- = 0$, $q_+ = \delta$ gives separation δ and $h_A = 0$. We therefore assume $w > 0$ in all the divisions below. First establish the following additive lower bound for $n \ge 1$ independent runs of A. If an additive output with bounded component functions $F = c + \sum_{i=1}^n Y_i$ has worst RMS $r < \rho/64$, put $b_i = \mathbb{E}_0 Y_i$, $b = \mathbb{E}_0 F$, and choose I uniformly in $\{1, \dots, n\}$ independently. Run just its component on one coin and return $Z = n(Y_I - b_I) + b$. This preserves every population mean and uses base cost w.

For $\theta = \theta(q)$, the laws of each component output under q and 0 have a common subprobability of mass $1 - \theta$: integrate the negative-sign kernel with weight $1 - q(d)$. For any two distributions sharing such a subprobability, writing each as a mixture of its normalized common part and residual part shows

$$(\mathbb{E}_q Y_i - \mathbb{E}_0 Y_i)^2 \le \frac{2\theta}{1-\theta}(\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i). \tag{C.7}$$

Indeed the squared displacement of either mixture mean from the common mean is at most $\theta/(1-\theta)$ times its variance; use $(u-v)^2 \le 2u^2 + 2v^2$. The case $\theta = 0$ is identical laws. Independence and $\sup_q \operatorname{Var}_q F \le r^2$ imply

$$\sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{4\rho}{1-\rho} r^2, \quad \operatorname{Var}_q Z \le n \sum_i \operatorname{Var}_q Y_i + n \sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{7}{3} n r^2.$$

All outputs here are bounded, and stopping is capped, so (C.BC) applies to Z without an additional integrability assumption. It gives $r^2 \ge 3c_b J/(7nw)$. Including the branch $r \ge \rho/64$ establishes

$$r \ge c_a \min\{\rho, \sqrt{J/(nw)}\}, \qquad c_a = \min\{1/64, \sqrt{3c_b/7}\}. \tag{C.8}$$

For $w > 0$, let S_n be the saddle quantity S from the proof of Lemma 40 for n independent copies of the finite statistic of A, on the parameter set $\mathcal{X}_\rho$. Equations (C.3) and (C.5) of that proof show

$$2S_n = \max\{\theta(q_+) - \theta(q_-) : \mathrm{Aff}(P_{A,q_+}, P_{A,q_-})^n \geq 1/2\}, \quad \inf_{F \text{ additive}} \sup_q \|F - \theta(q)\|_2 \leq 3S_n.$$

The latter constant is valid because $1 + 2/\log 2 + 2/(\log 2)^2 < 9$. The finite statistic outputs are bounded, so (C.8) applies. Hence, for $\beta = 2c_a/3$,

$$2S_n \geq \beta \min\{\rho, \sqrt{J/(nw)}\}. \tag{C.9}$$

Take $c_\delta = \beta/2$ and $C_H = 8/\beta^2$. For $0 < \delta \leq c_\delta \rho$ set $x = \beta^2 J/(4\delta^2 w)$. If $x \geq 2$, choose $n = \lfloor x \rfloor$, so $x/2 \leq n \leq x$. Equation (C.9) gives a pair of separation at least 2δ, with

$$h_A \leq 1 - e^{-(\log 2)/n} \leq \frac{\log 2}{n} \leq \frac{8 \log 2}{\beta^2} \frac{\delta^2 w}{J} \leq C_H \frac{\delta^2 w}{J}.$$

If $x < 2$, the pair $q_- = 0$, $q_+ = \delta$ belongs to $\mathcal{X}_\rho$, has separation δ, and $h_A \leq 1$ suffices because $C_H \delta^2 w/J > 1$. □

Lemma 42 (One pair hard for every capped policy)**.** *Assume (C.BC), and keep the universal constants of Lemma 41. For every $0 < \rho \leq 1/4$ and $0 < \delta \leq c_\delta \rho$, there exist $q_-, q_+ \in \mathcal{X}_\rho$, with $\theta(q_+) - \theta(q_-) \geq \delta$, such that simultaneously for every randomized capped single-coin policy A,*

$$h_A(q_-, q_+) \leq C_H \frac{\delta^2}{\rho L H} \mathbb{E}_0 T_A. \tag{C.10}$$

Each policy's seed and full stopped observable history are included. The constants and pair are chosen before the policy.

Proof. Let $\mathcal{K}_\delta = \{(q_-, q_+) \in \mathcal{X}_\rho^2 : \theta(q_+) - \theta(q_-) \geq \delta\}$. It is nonempty compact convex in the weak-star topology. Fix any finite list of capped policies $A_1, \ldots, A_k$ and put $w_i = \mathbb{E}_0 T_{A_i}$ and $\lambda = C_H \delta^2/(\rho L H)$. For a simplex vector $\pi \in \Delta_k$, define the observable mixture policy that draws $i \sim \pi$, reveals this index in its independent seed, and runs A_i. This is still capped, by the largest of the finitely many caps. Its base cost is $\sum_i \pi_i w_i$ and, since the revealed labels occupy disjoint transcript components, its squared Hellinger distance is exactly $\sum_i \pi_i h_{A_i}$. Lemma 41 therefore gives

$$\min_{z \in \mathcal{K}_\delta} \sum_i \pi_i (h_{A_i}(z) - \lambda w_i) \leq 0 \qquad (\pi \in \Delta_k).$$

For the finite simplex the displayed function is continuous affine in π and continuous convex in z, by the finite observation reduction. Sion's theorem [20] now gives

$$\min_{z \in \mathcal{K}_\delta} \max_{1 \leq i \leq k} (h_{A_i}(z) - \lambda w_i) = \max_{\pi \in \Delta_k} \min_{z \in \mathcal{K}_\delta} \sum_i \pi_i (h_{A_i}(z) - \lambda w_i) \leq 0.$$

Thus the closed subsets $\{z \in \mathcal{K}_\delta : h_A(z) \leq \lambda \mathbb{E}_0 T_A\}$ have the finite intersection property as A ranges over all capped policies. Compactness of $\mathcal{K}_\delta$ gives a point in their full intersection, proving (C.10). This uses only finite revealed mixtures and a finite simplex minimax theorem; no topology on a space of all policies, unjustified exchange of arbitrary infima, or unobserved-mixture affinity identity is needed. □

C.5 From one universal pair to every population algorithm

Lemma 43 (Finite-prefix midpoint decomposition)**.** *Suppose $q_-, q_+ \in \mathcal{X}_\rho$ have this property: every observable randomized single-coin policy with a deterministic finite cap obeys*

$$\mathsf{H}^2(P^A_{q_-}, P^A_{q_+}) \le \lambda \mathbb{E}_0 T_A, \tag{C.B}$$

where the transcript includes its independent seed and stopped observations, and $\mathsf{H}^2(P,Q) = 1 - \int \sqrt{dP\, dQ}$. Let $\bar{q} = (q_- + q_+)/2$. Then every population algorithm capped after K total flips obeys

$$\mathsf{H}^2(P^{\mathrm{pop}}_{q_-}, P^{\mathrm{pop}}_{q_+}) \le 2\lambda \mathbb{E}_{\bar{q}} T. \tag{C.C}$$

Proof. Condition first on the original independent seed. Pad the list of observed coins to K slots by their first-observation indices; an unobserved slot contributes the empty word of probability one. The policy is a finite binary decision tree, whose choices are determined by past outcomes and the fixed seed. For a leaf z, let $h_i(z)$ be the ordered word observed from slot i. Under a single fresh coin from q_r, write its word probability as $p_r(h)$, $r \in \{-,+\}$. Independence of the coin tapes gives leaf density

$$P_r(z) = \prod_{i=1}^{K} p_r(h_i(z)).$$

Interleaving changes the tree and the words, but not this factorization. Set $m_i = (p_-(h_i) + p_+(h_i))/2$ and $c_i = \sqrt{p_-(h_i)p_+(h_i)}$. Word probabilities are affine in q, so $m_i = p_{\bar{q}}(h_i)$. These quantities are positive; empty words cause no exception. Since $0 \le c_i \le m_i$, the elementary product inequality gives

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j. \tag{C.D}$$

The left side sums over leaves to $\mathsf{H}^2(P^{\mathrm{pop}}_-, P^{\mathrm{pop}}_+)$, because the midpoint experiment is normalized. The ith right side sums to the squared Hellinger distance between the two hybrid population experiments having slot i drawn from q_- or q_+ and all other slots independently drawn from $\bar{q}$. Those finite experiments are normalized too. Integrating the original common seed preserves this inequality: joint Hellinger distance with a revealed common seed is the integral of conditional Hellinger distances.

For each slot i, realize the hybrid experiment by a single-coin policy A_i. Simulate all other slots using independent artificial tapes from the fixed known reference population $\bar{q}$, and supply slot i by one genuine coin. If that slot is never requested, take no genuine observations. Include the original seed and all artificial tapes in the independent auxiliary seed. This is an observable single-coin policy capped at K genuine flips. It may use the fixed reference pair as design constants, but these do not depend on the genuine coin's unknown endpoint law. The universal quantifier in (C.B) covers this policy. No artificial tape is supplied to the original population algorithm.

The hybrid population transcript is a measurable function of this larger seed and the stopped genuine observations. Hellinger data processing and (C.B) therefore bound its distance by $\lambda \mathbb{E}_0 T_{A_i}$. Since $0 \le \bar{q} \le 1/2$, the one-coin latent distribution under $\bar{q}$ dominates one half of the all-negative distribution, with the magnitude law fixed. The same domination holds after adjoining the common auxiliary seed and taking any measurable nonnegative work functional. Hence

$$\mathbb{E}_0 T_{A_i} \le 2 \mathbb{E}_{\bar{q}} T_{A_i}.$$

When the genuine coin too is from $\bar{q}$, its expected charged work is exactly the work on slot i in the original midpoint experiment. Summing over the at most K slots yields (C.C).

All simulations are only devices in the lower-bound comparison; every genuine flip of the population algorithm is counted exactly once. □

Proof of the lower bound in Theorem 37. Take the universal constants c_δ, C_H from Lemma 42. Choose a fixed

$$0 < \gamma \le \min\{c_\delta, 1/(4\sqrt{C_H})\}, \qquad \delta = \gamma \min\{\rho, \sqrt{J/B}\}.$$

That lemma provides one pair $q_-, q_+ \in \mathcal{X}_\rho$ with target separation at least δ and property (C.B) with $\lambda = C_H\delta^2/J$. The pair is valid simultaneously for all finite caps and policies; it is not reselected after seeing a population transcript.

Let A be any full population algorithm with uniform expected work at most B. Apply Lemma 43 to its prefix capped after K flips, recording whether it already halted and, when relevant, its terminal output as a measurable statistic. Since $\bar{q} \in \mathcal{X}_\rho \subset \mathcal{Q}_\rho$, its midpoint expected capped work is at most B. Thus every prefix has squared Hellinger distance at most $2C_H\delta^2 B/J \le 1/8$. The inequality $\mathrm{TV}(P,Q) \le \sqrt{2\mathsf{H}^2(P,Q)}$ bounds its total variation by $1/2$.

Uniform expected work implies almost-sure termination separately under both endpoints. For any terminal event E, the event $E \cap \{T \le K\}$ is determined by the capped prefix, and the omitted endpoint probabilities are at most $\mathbb{P}_{q_-}(T > K) + \mathbb{P}_{q_+}(T > K) \to 0$. Consequently the terminal transcript's total variation is also at most $1/2$. This passage assumes neither a globally bounded number of coins nor differentiability or a chain rule at an unbounded stopping time.

If the estimator has infinite worst-case squared error, the lower bound is automatic. Otherwise threshold its output halfway between the endpoint targets. With endpoint separation $\Delta \ge \delta$ and worst RMS r, Markov's inequality bounds the sum of the two testing errors by $8r^2/\Delta^2$. Every test has error sum at least $1 - \mathrm{TV} \ge 1/2$. Hence $r \ge \Delta/4 \ge \delta/4$, which proves the lower inequality in (C.A), including saturation. □

C.6 Consequence for the proposed rare-event modulus

For this exact law the variational optimizer gives a cutoff $t \in (0, a)$:

$$\rho = (t/a)^2[1 + 2\log(a/t)], \qquad \Psi_{\mu,\rho}(h) = \begin{cases} [2ht(1 + \log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \ge t. \end{cases} \tag{C.E}$$

These are inherited variational identities, not an attaining statistical algorithm. Put $u = 1 + 2\log(a/t)$. Then $\rho = ue^{1-u}$ and $L = u - \log u$, so $L \le u \le 2L$. For $h < t$, $\Psi(h) = (ht/a^2)(u + 1 - h/t)$; its coefficient lies between u and $u + 1$. Together with the saturated branch this proves, with universal constants and fixed $a = 1/4$,

$$\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L/B}\}.$$

For instance take $\rho_n = \exp(-\exp n)$ and $B_n = \rho_n^{-2}$. Both radii are eventually unsaturated, and (C.A) implies

$$\frac{R_{\mu,\rho_n}(B_n)}{\Psi_{\mu,\rho_n}(B_n^{-1/2})} \ge c'\sqrt{1 + \log(1 + \log(1/\rho_n))} \longrightarrow \infty.$$

Thus no one finite universal C can satisfy the conjectured upper bound even at this single known law, with the algorithm knowing ρ and B. The missing statistical cost is not hidden magnitudes alone: known-profile submodels can behave differently. The lower bound uses the entire convex nuisance class to obtain a common hard pair. It does not assert that the radial variational optimizer itself is that pair, nor give a closed-form expression for the pair selected by compactness. No claim for every other magnitude law follows from (C.A).

C.7 Critical upper construction

Put $H = 1 + \log L$. For $j \geq 0$ choose

$$\varepsilon_j = 16^{-(j+1)}, \qquad N_j = \lceil (a^2 \rho \varepsilon_j)^{-1} \rceil, \qquad K_j = \lceil \log_2(a\sqrt{N_j}) \rceil.$$

For $0 \leq k \leq K_j$, let $r_k = a2^{-(k+1)}$ and, after n observations with h_n heads, define

$$L^+_{k,n} = (1 + 2r_k)^{h_n}(1 - 2r_k)^{n-h_n}, \qquad L^-_{k,n} = (1 - 2r_k)^{h_n}(1 + 2r_k)^{n-h_n}.$$

Before N_j, stop with output zero if a negative likelihood reaches $(K_j + 1)/\varepsilon_j$, or with output one if a positive likelihood reaches $(K_j + 1)/(\rho\varepsilon_j)$. Fix zero priority for ties. At N_j, if no earlier decision was made, return the strict majority bit. Denote this bounded, observable classifier by Z_j and its work by T_j.

For the wrong-sign likelihood, each next multiplier has mean $1 - 4r_k d \leq 1$. Stopping this nonnegative supermartingale at a crossing and a deterministic horizon, and then increasing the horizon, bounds the probability of crossing level b by $1/b$. Summing over $K_j + 1$ references gives wrong-crossing probabilities at most ε_j on a positive coin and $\rho\varepsilon_j$ on a negative coin. The wrong forced-majority event is a subset of the full-tape event, of probability at most $e^{-2N_j d^2}$. This is Hoeffding's bound: the centered Bernoulli log moment-generating function has second derivative at most $1/4$, so is at most $\lambda^2/8$; use independence and exponential Markov with $\lambda = \pm 4d$. For the exact power law,

$$\mathbb{E}_\mu e^{-2ND^2} = \frac{1 - e^{-2Na^2}}{2Na^2} \leq \frac{1}{2Na^2}.$$

No independence of a stopping event and this full-tape majority event is used. It follows, uniformly over the global-rarity class, that

$$\mathbb{P}(Z_j \neq S) \leq \tfrac{5}{2}\rho\varepsilon_j, \qquad \mathbb{P}(Z_j = 1, S = 0) \leq \tfrac{3}{2}\rho\varepsilon_j. \tag{C.U1}$$

For $d \geq N_j^{-1/2}$, choose $k = \lceil \log_2(a/d) \rceil \leq K_j$. Then $d/4 \leq r_k \leq d/2$, and the correct-sign log likelihood has mean increment at least $2d^2/3$: use

$$\tfrac{1}{2}\log(1 - 4r_k^2) + d \log \frac{1 + 2r_k}{1 - 2r_k} \geq -\tfrac{8}{3}r_k^2 + 4dr_k \geq \tfrac{2}{3}d^2.$$

Its upward increment is less than one. The bounded stopped-sum identity for the crossing time σ at a finite horizon n gives $g\mathbb{E}(\sigma \wedge n) \leq A_{j,s}$, where

$$A_{j,s} = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s \log(1/\rho).$$

Monotone convergence proves finite expected crossing time and, including the cap and the trivial small-gap case,

$$\mathbb{E}(T_j \mid d, s) \leq \min\{N_j, (3/2)A_{j,s}/d^2\}. \tag{C.U2}$$

Write $c_0 = 3/2$. Direct integration of the negative contribution gives

$$\mathbb{E} \min\{N, c_0 A/D^2\} \leq (c_0 A/a^2)[1 + \log_+(Na^2)].$$

For positives, layer cake and $\mathbb{E}[q\mathbf{1}_E] \leq \min\{\rho, \mu(C.E)\}$ give

$$\mathbb{E}[q \min\{N, c_0 A/D^2\}] \leq \int_0^\rho \min\{N, c_0 A/(a^2 u)\}\, du \leq (c_0 A/a^2)[1 + \log_+(Na^2\rho)].$$

The exact integral is $N\rho$ when $Na^2\rho \le c_0 A$, and otherwise is $(c_0A/a^2)[1+\log(Na^2\rho/(c_0A))]$. Thus this step also enforces the global, rather than pointwise, rarity condition.

Elementary bounds from the definitions are

$$N_j a^2 \le 2/(\rho\varepsilon_j), \qquad K_j + 1 \le 5(L+j+1),$$
$$A_{j,0} \le 7H(j+1), \qquad A_{j,1} \le 7H(j+1) + L,$$
$$1 + \log(N_j a^2) \le 5L(j+1), \qquad 1 + \log(N_j a^2 \rho) \le 5(j+1).$$

For example, $\log(L+j+1) \le \log L + j + 1$ and $4\log 2 < 3$ prove the bound on $A_{j,0}$. Substitution in (C.U2) and the two integrals gives

$$\mathbb{E}T_j \le (225/2)a^{-2}LH(j+1)^2. \tag{C.U3}$$

Independently choose a level J with $p_j = 2^{-(j+1)}$. For $J = 0$ return $Y = Z_0/p_0$ on one fresh coin. For $J = j \ge 1$, run Z_j and Z_{j-1} on successive unused blocks of that same coin, set $\Delta_j = Z_j - Z_{j-1}$, and return $Y = \Delta_j/p_j$. The bounded stopping times make the iid suffix argument elementary, as in the restart argument above; both decisions share the same latent sign. A disagreement implies an error of one classifier. Hence (C.U1) gives

$$\mathbb{E}Z_0^2 \le \tfrac{35}{32}\rho, \qquad \mathbb{E}\Delta_j^2 \le \tfrac{85}{32}\rho 16^{-j} \quad (j \ge 1).$$

Tonelli and the exact geometric sum now give

$$\mathbb{E}Y^2 \le \frac{35}{16}\rho + \frac{85}{16}\rho\sum_{j\ge1} 8^{-j} = \frac{165}{56}\rho.$$

The disagreement bounds also give absolute summability of expected increments, so signed expectations may be telescoped after integrability is established: $\mathbb{E}Y = \lim_j \mathbb{E}Z_j = \theta$. Work costs at most twice (C.U3), and $\sum_j p_j(j+1)^2 = 6$, yielding

$$\mathbb{E}T_Y \le 1350a^{-2}LH.$$

All operations are on one observable coin with independent seed; the random level is unbounded but the separate expectations are finite. This completes the upper checkpoint used in Theorem 37. Its actual work also has genuinely unbounded support: at any fixed observation count n, all likelihoods are at most $(1+a)^n$. For sufficiently large j, both the smallest stopping boundary and N_j exceed the relevant finite thresholds, so $T_j > n$ on every tape. Since $\mathbb{P}(J = j) > 0$, actual work exceeds every finite cap with positive probability while retaining finite mean.

Proof of the upper bound in Theorem 37. The preceding construction supplies a one-coin unbiased output with $\sup_q \mathbb{E}_q Y^2 \le (165/56)\rho$ and $\sup_q \mathbb{E}_q T_Y \le W_* = 1350a^{-2}LH$. Independently select $N \sim \text{Poisson}(B/W_*)$ and return $\widehat{\theta} = (W_*/B)\sum_{i=1}^N Y_i$ from independent genuine coins. Conditioning on N proves unbiasedness, uniform expected work at most B, and variance at most $(165/56)\rho W_*/B$. All moments are finite before using these identities, by the bounds and the finite Poisson moments. Choose the zero-work constant $\rho/2$ whenever its known error bound is smaller. Both choices are observable and depend only on known parameters. With a fixed, this proves the upper inequality in (C.A). This established replication device is not claimed as new. □

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